

Reassessed & Repossessed: Who Bears a Reappraisal Shock

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Abstract

North Carolina holds a county's assessed values fixed for up to eight years and then resets them to market in a single step. On a parcel-year panel of administrative tax bills covering 94.5 percent of Mecklenburg County's owner-occupied stock, I estimate a triple difference by synthetic difference in differences against thirty-three annually reappraising counties. The 2011 roll was regressive, assessing the cheapest quartile at 1.11 of sale price against 0.99 for the most expensive, and quarterly foreclosure filings among those homes rise 0.273 percentage points, with a 95 percent interval of 0.051 to 0.494. No other quartile moves. At the observed completion rate that is 423 households losing a home, 9.9 per thousand. A consumer address history matched to 96.8 percent of parcels follows them out. Bottom-quartile owners leaving after a filing left the county less often than their quartile peers leaving without one, 33.6 against 42.7 percent, and those who stayed landed in block groups 9 log points cheaper than the ones they left. Tenure falls with value: 14.2 percent bought their next home against 21.3, so of the 281 remaining in the county about 40 buy, 124 rent a house and 118 enter multifamily. The 2019 reset was level across the distribution and displaced no one. What repossesses is the regressive valuation, not the size of the bill.

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1 Introduction

Most American property tax systems do not revalue every year. Statutes set an outer bound on the interval and counties reappraise within it, so between one reappraisal and the next a home’s assessed value is frozen while its market value moves. If assessment tracks value one for one at each reset, then during the freeze the ratio of assessed to market value falls at exactly the rate the home appreciates. Homes that appreciate quickly become progressively under-taxed; homes that appreciate slowly, or fall in value, become relatively over-taxed. When the county finally reappraises, the whole accumulated divergence arrives in a single bill.

This paper asks who bears that deferred burden. The question matters because the answer is not a statement about assessors. A long cycle redistributes toward slow appreciators and away from fast ones during the freeze, then reverses the transfer at the reset, and it does so whether or not the appraisal is accurate. Which households sit on each side is an empirical matter that depends on where prices moved, and it can differ from one cycle to the next in the same county.

The literature has largely framed the distributional question differently, as one about error. Assessments are regressive with respect to sale price: assessors overvalue cheaper homes relative to expensive ones (McMillen and Singh, 2020a), a pattern linked to differences in appeal rates and transaction frequency (Amornsiripanitch, 2024; Avenancio-León and Howard, 2022). That literature studies the accuracy of a valuation at a point in time. The freeze is a different object, and in a county on a long cycle it can dominate: eight years of divergent appreciation moves the ratio of assessed to market value far more than any plausible appraisal error.

Setting. Mecklenburg County, North Carolina, contains Charlotte and reappraised in 2011 and 2019, the statute requiring a countywide reappraisal at least every eight years. Between those dates assessed values were frozen. The county’s assessor publishes a parcel-by-year record of administrative tax liabilities, which is unusual: most work in this area must impute the bill. I join it to ZTRAX assessor and transaction records, and sort homes into value quartiles on the median sale price of their census block group, placing 94.5 percent of the owner-occupied stock in a quartile.

What the freeze deferred. Between 2012 and 2018 the cheapest quartile of Mecklenburg neighbourhoods appreciated twice as fast as the most expensive. Since the assessed value was frozen throughout, that divergence is visible directly in the ratio of assessed value to sale price, which fell from 1.11 to 0.60 for the cheapest quartile and from 0.99 to 0.73 for the most expensive. The cheapest homes ended the freeze markedly *under*-assessed relative to the most expensive, which is the reverse of the pattern the regressivity literature describes. The 2019 reappraisal then raised their bills 28.2 percent against 6.8 for the most expensive. Section 5.1 shows that gradient to be a catch-up to appreciation already earned. The assessor is level

across the value distribution at the moment of the reset, and the reset recovers only 42 percent of what the freeze deferred.

What the settlement did. A single county experiencing a single countywide event cannot identify the level effect of that event. An indicator for the reappraisal year is an indicator for the calendar year, so any secular movement in the outcome loads onto it. What is identified is the contrast across quartiles, and that is also what the distributional question requires. Estimating it by synthetic difference in differences (Arkhangelsky et al., 2021) against counties that reappraise annually, foreclosure filings among the cheapest quartile rise 0.273 percentage points at the 2011 reappraisal and move -0.080 at the 2019 one. The two events settled the same kind of accumulated bill. What distinguished them was the valuation: the 2011 roll was regressive by the measure above and the 2019 roll was level. Distress followed the valuation error rather than the catch-up.

The magnitudes make that reading easier to accept. A bottom-quartile owner whose assessment caught up in 2019 pays about \$391 a year more, \$33 a month, against a home that appreciated 61 log points over the preceding six years. Capitalised at five percent the tax change is worth 2.6 percent of value. A settlement of that size, correctly assessed, is absorbed; the same settlement carrying a valuation error is not.

Contribution. The paper offers an administrative levy panel matched to a statutory shock, a measurement of what an eight-year freeze defers and to whom, and a way of separating the freeze from appraisal error that uses the arithmetic of the frozen ratio rather than an imputed market value. The reading for policy follows: the length of the cycle governs how large a settlement arrives, and the quality of the valuation governs whether it does damage. North Carolina moved Mecklenburg to a four-year cycle after 2019, which addresses the first.

2 Background

2.1 What a freeze defers

Write A_{it} for the assessed value of home i in year t and V_{it} for its market value. A county reappraising in year τ sets $A_{i\tau} = \rho_{i\tau} V_{i\tau}$, where ρ is the ratio the assessor achieves and is one under a statute requiring full-value assessment. Between reappraisals A is frozen while V moves, so for $t > \tau$,

$$\log \frac{A_{it}}{V_{it}} = \log \rho_{i\tau} - (\log V_{it} - \log V_{i\tau}). \quad (1)$$

The ratio of assessed to market value falls one for one with appreciation since the last reset. A home appreciating quickly becomes progressively under-taxed relative to what it is worth; a home appreciating

slowly becomes relatively over-taxed. None of this requires the assessor to err. It follows from holding A fixed while V moves.

At the next reappraisal A resets and the accumulated divergence arrives at once. Differencing (1) across consecutive resets in years τ and τ' ,

$$\Delta \log A_i = \underbrace{(\log V_{i\tau'} - \log V_{i\tau})}_{\text{appreciation over the freeze}} + \underbrace{(\log \rho_{i\tau'} - \log \rho_{i\tau})}_{\text{change in the assessor's aim}}. \quad (2)$$

Two things can therefore make one group's bill rise more than another's at a reappraisal, and they carry opposite readings. The group may have appreciated faster over the freeze, in which case the reset is a catch-up to value that had already accrued and the burden of a long cycle falls on fast appreciators. Or the assessor may treat the group differently at one reset than at the other, in which case the reset moves a group's tax relative to its value and the burden reflects appraisal error. The first says the policy lever is the length of the cycle; the second says it is the quality of the appraisal.

Equation (2) also shows how to separate the two without imputing a market value for unsold homes, which is what makes the question tractable here. The second term is the change in ρ measured *at each reset*, observable on homes that sell in a reappraisal year. The first is the drift in that same ratio during the freeze, and because A is frozen the drift is measurable for the whole stock: any parcel that sells at any point in the cycle reveals how far its assessment has fallen behind. Section 5.1 carries out both measurements.

Two features of the setting sharpen the design. Reappraisal timing is set by statute rather than chosen in response to any home's value, so which years are reset years is not a function of what happened to prices. And the assessed value is genuinely frozen: between reappraisals the bill moves only through the millage rate, which is set jurisdiction-wide and falls on every home in a jurisdiction alike, carrying no gradient in home value of its own.

2.2 Assessment regressivity

The distributional literature on the property tax has largely studied the second term in (2). A stylised fact is that assessment ratios are higher for lower-priced homes, so bills are regressive with respect to sale price (McMillen and Singh, 2020a). Assessed values are set below and rise less than one for one with sale prices, in part because wealthier owners appeal more often and mass-appraisal models fit sparse, low-turnover neighbourhoods poorly (Amornsiripanitch, 2024). The pattern is documented at both the county and the jurisdiction level (Amornsiripanitch, 2024; Avenancio-León and Howard, 2022), and McMillen and Singh (2020b) shows that homestead exemptions, though they cut liabilities more for lower-value homes, do not

remove it.

That literature measures the accuracy of a valuation at a point in time. The freeze is a different object, and in a county on a long cycle it is the larger one: eight years of divergent appreciation moves A/V further than any plausible appraisal error. The two are also distinguishable in principle, because regressivity is a property of a particular roll and can differ between two valuations produced by the same office, while the freeze operates in every cycle. Section 5.1 shows that in Mecklenburg the two point in opposite directions across the two cycles observed here, which is what allows them to be disentangled.

2.3 Mecklenburg County, North Carolina

The setting for my study is a fast-growing county with infrequent reappraisal cycles: Mecklenburg County, North Carolina. Reappraisal schedules are bounded by state law but set locally in most states, including North Carolina, where the statute requires a countywide reappraisal at least every eight years.¹ Mecklenburg County contains the city of Charlotte and six other jurisdictions, all subject to annually changing millage rates. Millage rates can therefore be raised in non-reappraisal years to lift revenue despite frozen assessed values.²

This setting creates useful variation in the tax burden. Because assessed values are frozen between reappraisals while sale prices continue to move, a home's assessment falls further behind what it would fetch with every year that passes, and the gap closes when the county reappraises. Which homes carry the largest gap by the end of a cycle is an empirical question about where prices moved, and Section 5.1 answers it for both of Mecklenburg's cycles.

3 Empirical Strategy

The freeze defers a burden and the reappraisal settles it. Measuring who bears that settlement requires two things the previous section has already supplied: a decomposition of the bill change into appreciation and appraisal error, and a comparison group for whom no settlement is ever due.

That comparison group is the reason this design uses counties that reassess annually. Where reappraisal is annual the assessed value tracks market value continuously, so no divergence accumulates unrecognised and no discrete correction is owed. Owners in such a county are exposed to the same regional housing and labour cycles as Mecklenburg owners and experience nothing at the Mecklenburg reappraisal dates. What

¹Mecklenburg moved to a four-year cycle after the 2019 reappraisal; the state mandate is unchanged as of 2026. Montana, Nebraska, North Dakota, Ohio, Oregon, Rhode Island and Tennessee also have reappraisal intervals extending into the six-to-ten-year range.

²Sub-county variation in millage would be a second channel for regressivity if cheaper homes sorted into jurisdictions raising rates more aggressively. Table A5 shows they do not.

separates the two is the settlement, which is the object of interest.

The comparison the paper draws is a triple difference: the cheapest quartile against the rest, Mecklenburg against annually reappraising counties, after the reappraisal against before. Section 3.1 states it as a single estimator. Section 3.2 explains why the comparison runs across counties rather than within Mecklenburg, and what the county’s own levy panel is used for instead.

3.1 Synthetic difference in differences

The comparison unit is built by synthetic difference in differences (Arkhangelsky et al., 2021), which nests the two estimators the design would otherwise choose between. Difference in differences weights every control county equally and every pre-period equally, and is credible only if the treated and control paths were already parallel. Synthetic control chooses county weights to reproduce the treated path but gives every pre-period equal weight and, in its original form, admits no intercept, so it is asked to match levels rather than trends. SDID does both jobs and adds a third: it chooses *time* weights.

The estimand. This paper’s contrast is a triple difference, so I place the quartile comparison inside the outcome and let SDID handle the county and time dimensions. For county i and quarter t ,

$$Y_{it} = \bar{y}_{it}^{Q1} - \frac{1}{3} \sum_{q=2}^4 \bar{y}_{it}^q, \quad (3)$$

the cheapest quartile’s outcome minus the mean of the other three. The estimator then solves

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min \sum_i \sum_t (Y_{it} - \mu - \alpha_i - \beta_t - W_{it} \tau)^2 \hat{\omega}_i \hat{\lambda}_t, \quad (4)$$

where W_{it} marks Mecklenburg after its reappraisal. Because the quartile contrast is already inside Y , the unit effect α_i absorbs a county’s average *gap* between cheap and expensive homes rather than its price level, and β_t absorbs the average movement of that gap across counties in a quarter. The identifying assumption is correspondingly weaker than a levels design would need: absent the reappraisal, Mecklenburg’s quartile gap would have moved parallel to the weighted donors’ quartile gaps. $\hat{\tau}$ is the triple difference.

The weights. Unit weights solve

$$(\hat{\omega}_0, \hat{\omega}) = \arg \min_{\omega_0 \in \mathbb{R}, \omega \in \Omega} \sum_{t \in \mathcal{T}^-} \left(\omega_0 + \sum_j \omega_j Y_{jt} - Y_{0t} \right)^2 + \zeta^2 T_{\text{pre}} \|\omega\|_2^2, \quad (5)$$

with Ω the simplex. The intercept ω_0 means the weights must make the donor path *parallel* to Mecklenburg’s, not identical to it, which is the right requirement when unit fixed effects will absorb any remaining

level difference. And the ridge penalty ζ , set as in Arkhangelsky et al. (2021) to $(N_{\text{tr}}T_{\text{post}})^{1/4}$ times the standard deviation of the donors' one-period changes, disperses the weights and makes them unique, whereas unpenalised fits could concentrate weights across just a couple counties.

Time weights solve the analogous problem across units,

$$(\hat{\lambda}_0, \hat{\lambda}) = \arg \min_{\lambda_0 \in \mathbb{R}, \lambda \in \Lambda} \sum_j \left(\lambda_0 + \sum_{t \in T^-} \lambda_t Y_{jt} - \frac{1}{T_{\text{post}}} \sum_{t \in T^+} Y_{jt} \right)^2, \quad (6)$$

choosing the pre-period quarters that predict the post-period average for the *donors*. This is the feature that matters most here. The 2011 window opens in 2007, and quarters in which every county in the country was in free fall carry little information about where a county was heading by 2013; difference in differences would weight them equally with the rest, and an event study would effectively give the last pre-quarter all the weight. SDID lets the data choose, and Figure 2 plots what it chose.

Inference. With one treated unit the jackknife is undefined and the bootstrap unreliable, as Arkhangelsky et al. (2021) note, so I use their placebo variance: each donor takes a turn receiving a pretend treatment, the whole estimator is re-run on the donors alone, and the dispersion of those placebo estimates is the variance. Its validity rests on the treated unit's noise resembling the donors', which Section 5.3 checks directly.

Donor eligibility. The pool is the set of counties satisfying the five conditions of Section 4, of which the first and most important is that the county reassess *annually*, so that it absorbs no discrete reappraisal of its own inside a Mecklenburg window. A control that is itself treated breaks the design and no choice of unit weights repairs it. Thirty-three counties qualify at the 2011 reappraisal and thirty-four at 2019.

3.2 Why the comparison runs across counties

The obvious alternative to a donor pool is to stay inside Mecklenburg and compare value quartiles to each other. The county's administrative levy panel makes that tempting, and it is worth saying why the paper does not do it.

A reappraisal-year indicator is not an instrument here. Reappraisal in North Carolina is countywide, so within one county an indicator for the reappraisal year is a function of the calendar year alone and carries no within-year variation of any kind. Dropping calendar-year fixed effects to make room for it does not create identifying variation; it conceals the absence of it. With reappraisal-cycle effects instead, the surviving contrast is the reappraisal year against the remaining years of its own cycle, and any secular movement in the outcome over those years is attributed to the reappraisal. The direction of the resulting bias is predictable from the raw series: mean log sale price in this county rises monotonically in every quartile

in every year of the 2019 cycle, and the foreclosure filing rate falls monotonically in every quartile in every year. A dummy for the first year of such a window recovers the cumulative trend over the remainder with the opposite sign. Prices trend up, so the indicator loads negative; filings trend down, so it loads positive.

Absorbing the trend does not rescue the design. One can put calendar-year effects back in and read the reappraisal off year-by-quartile interactions, which identifies the distributional contrast even though the level effect is lost. That specification is legitimate and I estimated it. It is also fragile in a way the donor-based estimator is not, because it must assume a counterfactual for the convergence visible in Mecklenburg’s own quartile gap. That gap falls monotonically through the panel as the financial crisis unwinds, from 1.31 percentage points in 2012 to 0.23 in 2017, a linear trend of -0.28 points a year. Normalising on 2018 sets the 2019 estimate at zero; normalising on 2015 gives -0.36 ; removing the fitted pre-trend gives $+0.40$. The answer is a choice, not a finding.

Donor counties supply what that design has to assume. Counties reappraising annually experienced the same post-crisis convergence and no reappraisal shock, so the weighted combination of Section 3.1 estimates the convergence rather than requiring it to be extrapolated. That is why the paper’s estimates come from across counties. Note also that difference in differences with uniform unit and time weights is one of the three estimators Table 9 reports, so the within-county comparison is nested in what is already shown.

What within-county variation is good for. One question the donor design cannot answer is whether the *size* of a household’s own bill increase is what matters. A county-level estimator sees only quartile averages, and the reappraisal’s correction varies a great deal within a quartile: among the cheapest homes in 2019 its standard deviation is 19.1 points around a mean of 28.3, because parcels differ in how stale their assessments had become. Section 5.7 uses that variation directly.

4 Data

The main source for this study comes from the Zillow Transaction and Assessment Database (ZTRAX, ICPSR Study 39652). The assessor database within ZTRAX contains a cross section of the universe of the U.S. housing stock as of 2024 and provides detail on key physical property characteristics and owner-occupancy status. I join it with a parcel key to the transaction database within ZTRAX which contains all transactions for each parcel since 1990 with detail on sales price, deed type, mortgage amounts. I express every dollar amount in 2025 dollars using the annual-average consumer price index for all urban consumers.³ All four outcomes are sourced from this file: the probability of transacting, the log sale price and log price per square foot on transacting parcel-years, and an indicator for a foreclosure filing in the parcel-year.

³<https://fred.stlouisfed.org/series/CPIAUCSL>

The resulting dataset links the assessor’s tax panel to transaction detail in each parcel-year. I sort homes into value quartiles on the median constant-dollar arm’s-length sale price of their *census block group* in an anchor window closing before either event window opens.

Sorting on an area rather than on a parcel’s own prior sale is deliberate, and it answers two problems with the obvious alternative. First, coverage: only about half of the county’s owner-occupied parcels transacted in the anchor window, so a parcel-level sort describes the homes that changed hands rather than the county’s housing stock. The block-group sort places 94.5 percent of owner-occupied parcels in a quartile. Second, measurement error: where a parcel-level sort is available it usually rests on a single transaction, which is a noisy draw around a home’s position in the distribution. Sorting on a noisy draw puts transitorily cheap homes in the bottom quartile and transitorily expensive ones in the top, so those groups revert toward the middle over the following decade by construction. Because the reappraisal correction *is* accumulated appreciation, that reversion would manufacture part of the very gradient the design measures. A block-group median is taken over roughly two hundred sales, so the transitory component averages out.

The cost is that a block group is a neighbourhood and not a house. Census block group explains 51 percent of the variance in log anchor value across Mecklenburg parcels, over 538 units at a median of about two hundred sold parcels each; census tract explains 45 percent over 238 units. The remaining variation is within-area and is discarded, so exposure is measured with error and estimated effects are attenuated toward zero. The estimand is correspondingly “homes in the cheapest quarter of neighbourhoods” rather than “the cheapest quarter of homes.” I report the block-group cut because it retains more of the value signal at finer resolution. Table 8 shows what the shock looks like under the tract cut and under two sorts that use no area price at all.

Whether the choice matters for the results is itself testable, and Section 5.2 runs that test. The reappraisal gradient does not attenuate under the area sort; it is at least as steep as under the parcel-level cut, which is the opposite of what measurement error in exposure predicts and suggests that assessment staleness in this county is clustered in space rather than idiosyncratic across homes.

For Mecklenburg, I join the ZTRAX data with a panel of property tax bills for 2011 through 2022 from Mecklenburg County’s Open Mapping Tax Parcel CAMA Data.⁴ The records are a parcel-by-year panel of the county assessor’s administrative records and provide geographic identifiers including mailing address, assessor neighbourhood and tax district codes. This panel supplies the bill changes used in Section 5.2 and the dose-response test of Section 5.7.

Section 5.6 asks where households went after leaving a Mecklenburg home, and for that I add a consumer address history compiled by Data Axle and distributed through Dewey. It holds one record per adult and up

⁴<https://mecksusftp001.blob.core.windows.net/gis/OpenMapping/index.html>

to ten addresses for that person, each with parsed street components, a county code and the month it took effect, so a person's residential sequence is recoverable from a single record. The file is organised by current address rather than by history, which means a household that left Mecklenburg sits in whichever part covers where it went; recovering everyone who was ever in the county requires reading the whole national file, and 4.2 million people have a Mecklenburg address somewhere in their history. I match those addresses to parcels on postal code, house number and street name.

Table 1 describes the resulting estimation sample, quartile by quartile, and sets Mecklenburg beside the synthetic county the estimator builds for it. The quartiles are cut within each county on constant-dollar sale prices, so they partition its own stock rather than a national distribution, and the cheapest quartile is the group whose assessment had fallen furthest behind market value by the end of the freeze.

Table 1: Pre-assessment balance, Mecklenburg and its synthetic counterpart

	Mecklenburg	Synthetic Mecklenburg	Donor pool average
<i>Q1 (cheapest)</i>			
Transaction rate (%)	1.807	1.311	1.024
Foreclosure filing rate (%)	0.367	0.637	0.955
Log sale price	11.935	11.815	11.907
Log price per square foot	4.668	4.606	4.649
Sales per quarter	613	139	208
Parcels	33,903	9,478	19,725
<i>Q2</i>			
Transaction rate (%)	1.729	1.199	1.116
Foreclosure filing rate (%)	0.331	0.543	1.231
Log sale price	12.261	11.995	12.110
Log price per square foot	4.813	4.708	4.801
Sales per quarter	586	121	237
Parcels	33,902	9,529	19,718
<i>Q3</i>			
Transaction rate (%)	1.469	1.093	1.169
Foreclosure filing rate (%)	0.220	0.377	1.280
Log sale price	12.645	12.355	12.398
Log price per square foot	4.942	4.852	4.919
Sales per quarter	498	106	253
Parcels	33,906	9,536	19,711
<i>Q4 (most expensive)</i>			
Transaction rate (%)	1.526	1.084	1.240
Foreclosure filing rate (%)	0.153	0.266	1.177
Log sale price	13.199	12.815	12.928
Log price per square foot	5.160	5.032	5.150
Sales per quarter	517	102	268
Parcels	33,899	9,535	19,715

Notes. Means over the four years preceding the 2011 reappraisal, by value quartile, for Mecklenburg, for the weighted combination of donor counties the estimator selects, and for the unweighted average of the counties eligible to serve. Quartiles are cut within each county on constant-dollar sale prices, so Q1 is the cheapest quarter of that county's own stock. The weights are those fitted for foreclosure filings at the 2011 reappraisal. Weights are refitted for every outcome and each side of the contrast, so this is one of ten comparison units the design builds and not a balance table for the design as a whole; the cheapest quartile is built from the weights for that quartile and the remaining three from the weights for the pooled remainder. Because the estimator matches demeaned trajectories rather than levels, no row here is a restatement of what the fit was asked to do, and the synthetic county reproduces Mecklenburg's pre-reappraisal level of an outcome only insofar as matching its path also matched its level. The comparison with the unweighted pool shows what the weights buy. Levels differ out of the triple difference, which is what makes a difference in level between the treated county and its synthetic counterpart admissible rather than disqualifying.

Selection into the comparison unit. Every county with identifiable owner-occupants and a complete transaction record enters the candidate set. Which of them can serve as a control is then decided by five conditions. First, the county must assess annually, so it absorbs no discrete correction of its own inside a Mecklenburg window. Second, its deed recordings must credibly track its housing market rather than fill in over time.⁵ Third, the homes that received a value quartile must resemble those that did not. A parcel is

⁵Coverage in several states improves across the sample for administrative reasons, which produces a rising sales series that has nothing to do with housing. Both parts are required: counties whose sales fell during the crisis show low volume too, and

sorted only if it sold at arm’s length, with a recorded price, in the anchor window. If the homes that sold differ systematically from those that did not, the quartiles describe the homes that changed hands rather than the county’s stock, and the cheapest quartile is not its cheapest quarter. Within each county I compare sorted and unsorted parcels on assessed value, improvement value and square footage, and require the largest standardized difference to fall below 0.25.⁶ Counties sorting less than a tenth of their stock are also excluded, since the quartile cutpoints then rest on too few sales to be stable. Fourth, each quartile-year cell inside the window must hold enough sales to compute an outcome. Fifth, owner-occupants must be identifiable, since a universe containing rentals and second homes is not the universe the design compares.

Nine states assess annually, containing 745 counties, and I extract all of them. Ninety have no usable record of sales, leaving 655. A further 296 have no source from which owner-occupants can be identified, leaving 359. Seventy cannot be sorted into value quartiles, most because too few of their homes sold in the anchor window, leaving 289. Of these, 188 sort a stock that does not resemble the stock they leave unsorted, leaving 101 counties, or 13.6 percent of those an annual assessment regime makes available in principle. The two remaining conditions, coverage stability and cell size, depend on the assessment being studied and are reported with the results.

The screen governs the placebo distribution as well as the donor pool. Inference compares the treated county’s post-reappraisal divergence to the divergence of each donor treated as if it had been assessed. A county whose deed record is filling in shows a compressed pre-reappraisal path and a rising post-reappraisal one, so it registers a large apparent effect for reasons unrelated to any assessment and displaces Mecklenburg in the ranking. Excluding such a county from the weights while retaining it as a placebo would leave the comparison distorted.

The donor pool admits fifty-six counties at the 2011 reappraisal, listed with the value each condition took in Table A3, but the estimator does not use them equally. Weights are fitted separately for every outcome, assessment and side of the contrast, so each cell of the results has its own comparison group, and Tables 2 and 3 report which counties those are.

Three features govern how the estimates should be read. The weights are concentrated: the 2011 cells draw on four to eleven donors each, with the largest single county carrying between a quarter and half of the fit, so several estimates rest substantially on one or two comparison counties. The composition turns over between assessments: twenty donors receive weight somewhere in 2011 and twenty-nine in 2019, with

they are kept.

⁶The standardized difference is the conventional measure of comparability between two groups (Rosenbaum and Rubin, 1985), and a quarter of a standard deviation the conventional point beyond which the difference is treated as material (Imbens and Wooldridge, 2009). Mecklenburg sorts 39.2 percent of its stock with a largest standardized difference of 0.077, so the treated county is well inside it. Across the counties measured, imbalance is flat above a tenth coverage and rises sharply below it, which is where the floor is set.

seventeen common to both, which follows from solving (5) cell by cell rather than imposing one weight vector across outcomes. And participation is broad even where weight is not: only one county in the pool receives no weight in any cell, so the concentration reflects which counties fit best rather than a pool most of whose members are unusable.

Two consequences follow. Where an estimate rests mostly on one county, it is a different claim from one resting on many, and the tables let a reader see which they are looking at. And where two cells point the same way, they do so from separately chosen comparison groups, so the agreement is corroboration rather than a shared construction reproducing itself.

Table 2: Donor composition of synthetic Mecklenburg, by outcome and assessment.

Donor	Log sale price 2011	Log sale price 2019	Log price/sq. ft. 2011	Log price/sq. ft. 2019	Foreclosure filing 2011	Transaction 2011	Transaction 2019	Mean
Citrus FL	7.3	—	10.0	—	—	46.5	—	9.1
Arlington VA	10.8	—	17.3	—	—	—	31.7	8.6
Highlands FL	12.5	—	9.7	—	—	—	33.4	7.9
Jefferson KY	—	1.2	—	0.7	—	49.4	—	7.3
Hillsborough FL	—	12.8	—	30.6	—	—	—	6.2
Shelby AL	11.3	—	14.4	—	11.6	—	4.7	6.0
Duval FL	—	17.7	—	20.1	—	—	—	5.4
Cobb GA	—	—	—	—	36.1	—	—	5.2
Escambia FL	—	8.3	14.1	11.7	—	—	—	4.9
Ingham MI	1.0	—	—	4.3	28.2	—	—	4.8
Santa Rosa FL	10.7	—	14.3	—	—	—	—	3.6
County 25013 MA	—	—	—	—	24.1	—	—	3.4
Wayne MI	—	12.5	—	7.5	—	—	—	2.9
St. Lucie FL	13.5	3.0	—	—	—	—	—	2.4
51510 VA	—	—	—	—	—	—	13.9	2.0
Yavapai AZ	—	0.6	—	—	—	—	13.0	1.9
Clay FL	—	—	12.9	—	—	—	—	1.8
Kent MI	12.6	—	—	—	—	—	—	1.8
Oakland MI	—	8.9	—	3.6	—	—	—	1.8
26005 MI	—	9.6	—	1.7	—	—	—	1.6
Polk FL	—	7.6	—	1.3	—	—	—	1.3
Osceola FL	—	8.7	—	—	—	—	—	1.2
Washington MN	—	—	—	8.1	—	—	—	1.2
Orange FL	—	6.1	—	—	—	—	—	0.9
Ottawa MI	5.5	—	—	—	—	—	—	0.8
Mobile AL	2.9	—	2.4	—	—	—	—	0.8
Hernando FL	—	3.0	2.0	—	—	—	—	0.7
Pima AZ	—	—	2.5	1.8	—	—	—	0.6
Loudoun VA	4.3	—	—	—	—	—	—	0.6
Alachua FL	—	—	—	—	—	4.1	—	0.6
Seminole FL	—	—	—	3.7	—	—	—	0.5
Kalamazoo MI	3.4	—	—	—	—	—	—	0.5
Flagler FL	3.2	—	—	—	—	—	—	0.5
Dakota MN	—	—	—	2.9	—	—	—	0.4
Walton FL	—	—	—	—	—	—	2.7	0.4
Collier FL	—	—	—	1.7	—	—	—	0.2
Okaloosa FL	1.0	—	—	—	—	—	—	0.1
Prince William VA	—	—	—	—	—	—	0.5	0.1
25 further donors	—	—	—	—	—	—	—	—

Notes. Weights in percent, from (5), for the cells that pass the feasibility screen; cells the estimator cannot fit are omitted because a composition without a corresponding estimate invites over-reading. Weights below half a percent are shown as dashes, the optimizer returning small positive values rather than exact zeros. The final row collects donors receiving no weight in any cell; they are counted rather than dropped, since the share of the pool that contributes nothing is itself a measure of how concentrated the comparison is. The mean column averages across the five cells and is a summary of participation, not a weight used in any estimate.

Table 3: Donor composition for quartiles two to four.

Donor	Log sale price 2011	Log sale price 2019	Log price/sq. ft. 2011	Log price/sq. ft. 2019	Foreclosure filing 2011	Transaction 2011	Transaction 2019	Mean
Alachua FL	—	—	—	3.5	—	91.6	—	13.6
Chesterfield VA	41.1	—	52.8	—	—	—	—	13.4
Arlington VA	23.5	—	11.5	—	—	—	36.9	10.3
Shelby AL	—	4.2	—	7.6	25.1	—	18.4	7.9
Ingham MI	—	5.5	—	8.5	10.8	—	29.0	7.7
Fairfax VA	—	21.3	11.9	13.8	—	—	—	6.7
County 25013 MA	13.2	—	10.2	—	19.9	—	—	6.2
Cobb GA	—	—	—	—	29.9	5.2	—	5.0
Kalamazoo MI	16.6	8.4	—	—	—	—	—	3.6
Scott MN	—	10.4	—	5.2	—	—	—	2.2
Polk FL	—	6.0	—	8.6	—	—	—	2.1
Macomb MI	—	—	—	—	14.3	—	—	2.0
County 25025 MA	5.6	—	6.0	—	—	—	—	1.7
Hernando FL	—	6.2	—	5.0	—	—	—	1.6
Charlotte FL	—	5.9	—	4.9	—	—	—	1.5
Yavapai AZ	—	—	—	—	—	—	9.5	1.4
Wayne MI	—	—	—	8.9	—	—	—	1.3
Miami-Dade FL	—	8.2	—	—	—	—	—	1.2
Highlands FL	—	6.9	—	0.9	—	—	—	1.1
Walton FL	—	3.6	—	3.9	—	—	—	1.1
Jefferson AL	—	4.0	—	3.3	—	—	—	1.0
Indian River FL	—	2.9	4.1	—	—	—	—	1.0
Clay FL	—	2.0	—	4.5	—	—	—	0.9
Pinal AZ	—	—	—	1.9	—	—	2.6	0.6
Escambia FL	—	—	—	1.2	—	3.2	—	0.6
51510 VA	—	—	—	—	—	—	3.5	0.5
Dakota MN	—	—	—	3.5	—	—	—	0.5
Prince William VA	—	—	—	3.4	—	—	—	0.5
Collier FL	—	—	—	2.5	—	—	—	0.4
Mobile AL	—	—	2.5	—	—	—	—	0.4
Palm Beach FL	—	2.4	—	—	—	—	—	0.3
Duval FL	—	—	—	2.2	—	—	—	0.3
Maricopa AZ	—	—	—	2.1	—	—	—	0.3
Okaloosa FL	—	2.1	—	—	—	—	—	0.3
Washington MN	—	—	—	1.8	—	—	—	0.3
County 25017 MA	—	—	1.1	—	—	—	—	0.2
26005 MI	—	—	—	0.8	—	—	—	0.1
Oakland MI	—	—	—	0.7	—	—	—	0.1
Manatee FL	—	—	—	0.7	—	—	—	0.1
<i>24 further donors</i>	—	—	—	—	—	—	—	—

Notes. As Table 2, for the pooled remainder of the distribution rather than the cheapest quartile. Weights are fitted separately for the two sides of the contrast, so the counties reproducing Mecklenburg's bottom quartile need not be those reproducing the rest of its stock, and in several cells they are not.

4.1 The assessor's aim, and how far it slips

Figure 1 plots the assessor's value against the sale price for homes assessed and sold in the same year, pooling both reappraisals. The fitted slope is 0.832, below one, and the 45-degree line is the ratio the statute asks for. Read on its own this figure is the regressivity result: assessed value rises less than one for one with sale price.

Two qualifications matter for what follows, and Section 5.1 develops both. The pooled slope averages two rolls that differ. Split by cycle in Appendix A.6, it is 0.777 in 2011 and 0.905 in 2019: the later valuation tracks sale prices far more closely. And a slope estimated on all assessed-and-sold pairs mixes the assessor's aim with the drift that accumulates after it, because most such pairs come from years in which the roll was already stale. Isolating the aim requires reading the ratio in the reset year alone, which is what Table 5 does.

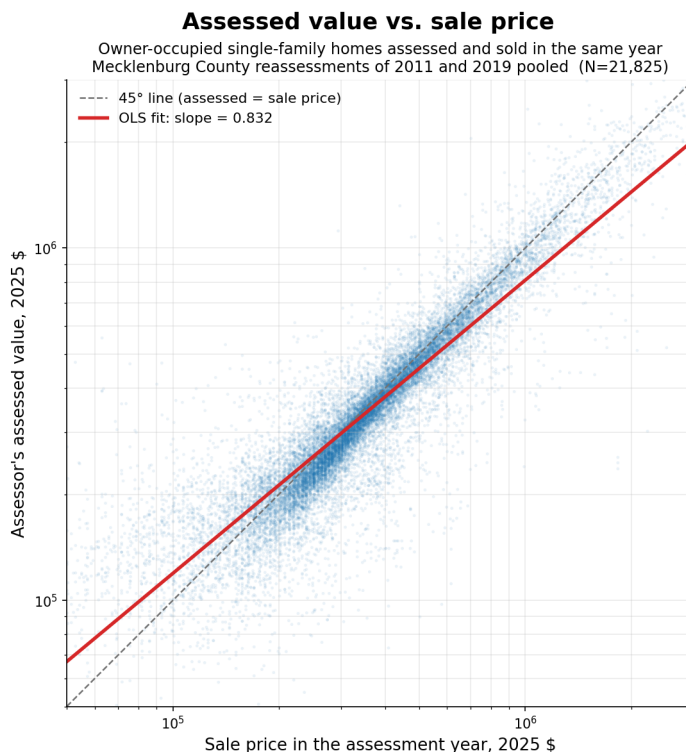


Figure 1: Assessed vs. sale-based value, Mecklenburg County for owner-occupied homes. The assessor's value against a sale-anchored market value; the below-one slope is the regressivity of the assessment.

Next, I document variation in the annual change in the property tax bill across home-value quartiles. This measure is inclusive of all property tax components, such as homestead exemptions and millage rate differences, because it is measured from the final bill the owner owes. Table 4 displays this variation for the four quartiles of Mecklenburg homes. The reappraisal year is the only year in which the four quartiles part company.

Table 4: Change in the property tax bill (%) by home-value quartile, Mecklenburg, 2012–2022.

	Millage (per \$100)	Change in tax bill (%)			
		Q1 (cheapest)	Q2	Q3	Q4 (most expensive)
2012	1.178	-2.30	-2.25	-2.21	-2.18
2013	1.205	1.95	2.03	2.03	1.55
2014	1.225	-0.03	1.48	1.49	1.83
2015	1.225	0.88	0.77	0.77	0.81
2016	1.232	0.28	0.20	0.32	0.46
2017	1.231	0.37	0.25	0.33	0.43
2018	1.246	1.31	1.26	1.23	1.20
2019	0.972	27.33	19.96	12.54	6.24
2020	0.977	1.45	1.03	0.74	0.78
2021	0.979	-0.25	0.04	0.61	0.96
2022	0.924	-4.29	-4.96	-3.95	-4.36

Notes. Mean year-over-year change in the property tax bill, in percent, for owner-occupied Mecklenburg homes by value quartile (Q1 cheapest). The bill is the county’s administrative levy, and the change is nominal: the bill is a nominal claim on the owner’s cash, and inflation is common to all parcels within a year, so it shifts every quartile equally and cancels out of the comparison across quartiles. Quartiles are cut on the median constant-dollar sale price of a parcel’s census block group in the anchor window, so a parcel need not have transacted to be sorted. Millage is the effective rate borne per \$100 of assessed value, the levy divided by the assessed value. The window opens in 2012 because the change requires a prior year. The horizontal rule marks the 2019 countywide reappraisal; the prior cycle began in 2011. Between assessments the bill still moves, because the millage rate is set annually even though the assessed value is frozen: the county-wide change runs from -5.0 to $+2.0$ percent across those years. What the reappraisal adds is not movement but a gradient: the change in the reappraisal year runs from 27.3 percent in the cheapest quartile to 6.2 in the most expensive, where in every other year the four quartiles move within a fraction of a point of one another. Changes are winsorized at the 1st and 99th percentiles.

5 Results

5.1 What the freeze deferred

Table 5 traces the assessed-to-sale ratio through both cycles and is the paper’s central descriptive fact. In 2012, a year into the freeze that followed the 2011 reappraisal, the cheapest quartile was assessed at 1.11 of sale price and the most expensive at 0.99. By 2018, the last year before the next reset, those had fallen to 0.60 and 0.73. The ranking has inverted. The cheapest homes end the freeze markedly *under*-assessed relative to the most expensive, which is the reverse of the pattern the regressivity literature describes and cannot be produced by an assessor who never revalued anything in between.

Equation (1) says what the drift measures. With A frozen, a fall in A/V is appreciation, one for one, and it is observable for any parcel that sells at any point in the cycle rather than only for parcels that sell twice. The cheapest quartile’s ratio fell 61 log points between 2012 and 2018 and the most expensive quartile’s 31.

Table 5: The assessed-to-sale ratio through the freeze.

	Q1 (cheapest)	Q2	Q3	Q4 (most expensive)	Q1 – Q4
2012 (reset)	1.113	1.073	1.012	0.992	+0.121
2013	0.993	0.983	0.927	0.930	+0.063
2014	0.948	0.942	0.901	0.868	+0.080
2015	0.873	0.866	0.848	0.832	+0.042
2016	0.774	0.801	0.786	0.804	-0.030
2017	0.667	0.725	0.734	0.759	-0.092
2018	0.603	0.661	0.688	0.731	-0.128
2019 (reset)	0.892	0.922	0.937	0.951	-0.059
2020	0.819	0.856	0.880	0.897	-0.078

Notes. Median administrative assessed value divided by sale price, owner-occupied Mecklenburg parcels selling in that year, both in 2025 dollars. North Carolina assesses at 100 percent of appraised market value, so the ratio is directly interpretable and needs no scaling. Reappraisals took effect in 2011 and 2019; the 2011 roll is observed from 2012 because the tax-change panel requires a prior year, so the figure in that row already includes one year of drift. Rows between the rules are years in which the assessed value was frozen and only the ratio’s denominator moved. Ratios outside $[0.2, 3]$ are dropped as data errors.

Is the reset a catch-up or an imposed error? Equation (2) localises the answer in the ratio measured at each reset. If the assessor is regressive, cheap homes carry a higher ρ in the year the roll is struck. If the reappraisal is a catch-up, ρ is level across quartiles at the reset and the entire bill gradient lives in the drift.

One correction is needed first. Assessed value is set as of the first of January while sales record throughout the year, and the median 2019 sale records 56 percent of the way through it. A fast-appreciating home has moved further from its January value by the time it sells, which depresses its ratio for a reason that has nothing to do with the assessor. I therefore discount each 2019 sale price back to the first of January at its own quartile’s annual growth rate, using the recorded date of the deed rather than assuming a mid-year sale.

Table 6: Where the reappraisal shock comes from.

	Q1 (cheapest)	Q2	Q3	Q4 (most expensive)	Q1 – Q4
Annual price growth, repeat sales (%)	8.24	7.03	4.89	3.38	4.86
parcels	151	390	567	613	
Price growth 2012–2018, whole stock (log pts)	61.4	48.5	38.5	30.6	30.8
Assessed-value catch-up at 2019 (log pts)	39.2	33.3	30.8	26.4	12.9
Tax bill change at 2019 (log pts)	24.9	18.6	12.1	6.6	18.3

Notes. Repeat sales compare the same parcel sold in 2012–2015 and again in 2019–2021, and never use an assessed value; the median annualised log growth is reported with the number of parcels contributing. Price growth over 2012–2018 is implied by the drift in the assessed-to-sale ratio through the freeze, following equation (1), and covers the whole stock rather than repeat sellers. The assessed-value catch-up is the change in that ratio between 2018 and 2019. The tax bill change is the mean nominal change in the administrative levy in 2019, in log points for comparability with the rows above.

So corrected, the 2019 reset leaves the cheapest quartile at a ratio of 0.938 and the most expensive at

0.964, a gap of 0.026 where the raw figure is 0.059. The reset is level across the value distribution to within three points of ratio, and what departure remains runs against regressivity rather than with it. The 2011 roll, by the same measure, left the cheapest quartile at 1.11 against 0.99: regressive, and that figure is a lower bound, because it is observed a year into the freeze and a year of faster appreciation at the bottom has already pulled the cheapest quartile's ratio down.

The catch-up is also incomplete. Table 6 sets 12.9 log points of differential assessed-value correction against 30.8 points of differential appreciation: the 2019 reappraisal closes 42 percent of the gap the freeze opened. The cheapest quartile leaves the reset still under-assessed relative to the most expensive, which is why the ratio series resumes its divergence immediately afterwards.

Three conclusions follow, and they organise the rest of the paper. The gradient in the 2019 bill is a catch-up to appreciation that had already occurred, not a tax imposed above value. The burden of an eight-year freeze falls on whoever appreciated fastest, which in Charlotte between 2012 and 2019 was the bottom of the market, and would fall elsewhere in a county where prices moved differently. And regressivity, present in the 2011 roll and absent from the 2019 one, is a property of a particular valuation rather than of the regime, so the two mechanisms are separable in this setting.

5.2 The shock the reappraisal delivers

Table 7 reports what the deferred correction is worth in a bill. In the reappraisal year the cheapest quartile's bill rises 28.2 percent and the most expensive quartile's 6.8, a spread of 21.5 percentage points. Averaged over every other year in the estimation window the same four quartiles rise between 0.41 and 0.49 percent, a spread of five hundredths of a point. What a reappraisal adds is not movement in the bill, which the annually reset millage rate supplies in both directions, but a gradient in that movement across the value distribution.

Table 7: The reappraisal shock by home-value quartile.

	Reappraisal year (2019)	Every other year (2012–2020)	Parcels
Q1 (cheapest)	28.25	0.49	42,381
Q2	20.43	0.42	49,187
Q3	12.82	0.41	52,255
Q4 (most expensive)	6.77	0.44	53,815
Q1 – Q4	21.48	0.05	

Notes. Mean year-over-year percent change in the administrative tax bill, owner-occupied Mecklenburg parcels, 2012–2020. Quartiles are cut on the median sale price of a parcel’s census block group in the anchor window, so a parcel need not have transacted to be sorted. Changes are nominal and winsorised at the 1st and 99th percentiles. To the extent the change comes from the millage rate it is common to all parcels within a year, which is why the four quartiles move together in every year but the reappraisal.

Table 8 asks whether that gradient is a property of the county or of how homes were sorted, and it answers a prior question at the same time: which sorting variable is the right one. Four are compared. The parcel’s own anchor sale price is home-level but available for two homes in five and rests on a single transaction, so it carries that draw’s transitory component; sorting on a noisy measure places transitorily cheap homes in the bottom quartile and dear ones in the top, and they revert toward the middle afterwards, which would manufacture part of a gradient built on accumulated appreciation. The block-group median averages that noise away over roughly two hundred sales and covers the whole stock, but measures a neighbourhood. A hedonic predicted value—log anchor price fitted on living area, rooms and age with block-group fixed effects, then predicted for every parcel—is the conditional expectation of value and so is free of transitory noise by construction, while retaining the within-neighbourhood variation the block-group median discards. The 2018 assessed value is predetermined with respect to the 2019 shock and covers the stock, but is the assessor’s own number.

Argument alone does not rank these, so I rank them on a test none was fitted to: how well each predicts the *actual* 2019 sale price of homes that sold that year. Every sort is built from data ending in 2006 or from the 2018 roll, so 2019 prices are out of sample for all four. The 2018 assessed value wins decisively, explaining 82 percent of the variance in the log 2019 sale price against 62 for the hedonic and 45 for either price-based sort. The assessor is stale in *level* by seven years and still ranks homes better than a decade-old transaction does, which is what one would expect of a measure built by visiting every parcel rather than by observing the minority that sold. Notably the block-group median matches the parcel’s own sale price on this test, 0.452 against 0.450, while covering 98 percent of the stock against 41: moving from the parcel to its neighbourhood costs no signal at all and doubles the sample.

Table 8: Four sorting variables, ranked out of sample and compared on every result.

Sorting variable	Coverage	R^2 on	2019 shock	Ratio gap	Ratio gap	2019 gradient, Q1–Q4	
	of stock	2019 price	Q1–Q4 (pp)	2018	at reset	Filing (pp)	Price (%)
Own anchor sale price	41%	0.450	+22.7	-0.114	-0.067	-0.090	-0.91
Block-group median (used)	98%	0.452	+22.4	-0.128	-0.059	-0.028	-0.69
Hedonic predicted value	98%	0.622	+23.2	-0.132	-0.075	-0.010	+0.10
2018 assessed value	63%	0.820	+25.3	-0.147	-0.080	-0.043	+2.52

Notes. Coverage is the share of parcels with a geography receiving a quartile. R^2 is from a regression of the log 2019 sale price on the log sorting variable, among the 8,553 homes that sold in 2019; no sort uses a 2019 transaction, so the test is out of sample for all four. The remaining columns hold the estimation sample fixed and swap only the sorting variable, so any difference is the sort and not the sample. The ratio gaps are the cheapest quartile’s median assessed-to-sale ratio minus the most expensive quartile’s, in 2018 and in the reappraisal year. The gradient columns are the 2019 movement of the cheapest quartile against the most expensive, relative to 2018, from a difference in differences on those two quartiles, 2019 against 2018, with the controls of Section efsec:synth.

Every conclusion in this paper survives every sort, and the two that matter strengthen as the sorting variable improves. The 2019 shock spread runs from 22.4 to 25.3 percentage points, widest under the sort that measures value best. The assessed-to-sale gap at the end of the freeze is negative under all four, between -0.11 and -0.15 , so the finding that cheap homes end the freeze *under*-assessed is not a construction of the block-group cut. The gap at the reset is likewise negative under all four, between -0.06 and -0.08 , so the reset is nowhere regressive. And the outcome gradients are indistinguishable from zero under all four, for both foreclosure filings and sale prices.

That the ranking of sorts tracks the strength of the freeze result is itself evidence. Measurement error in the sorting variable attenuates a true gradient toward zero, so if the freeze mechanism is real the gradient should sharpen as the sort improves, and it does: 22.4 points under the noisiest sort and 25.3 under the best. A gradient manufactured by mean reversion in a noisy sort would move the other way.

I report the block-group cut in the main results despite the assessed value winning the out-of-sample test, for one reason. Sorting on the assessor’s own number to study the assessor’s own behaviour invites a circularity: if the 2011 roll was regressive, and Section 5.1 finds it was, then a sort on the 2018 assessed value inherits that error. The sort uses only ranks, and regressivity would have to reorder homes rather than merely rescale them to bite, so the concern is limited. But it is not nothing, and the block-group median is free of it while costing no out-of-sample accuracy relative to the price-based alternative it replaces. Readers preferring the sharper instrument will find every result stronger under it.

5.3 Synthetic difference-in-differences estimates

Table 9 reports $\hat{\tau}$ from (4) for both reappraisals and five outcomes, with the difference-in-differences and synthetic control estimates alongside as in Arkhangelsky et al. (2021). The two reappraisals behave differently, and the difference is the paper’s most interesting result.

At the **2011** reappraisal the cheapest quartile’s foreclosure filing rate rises by 0.273 percentage points relative to the rest of the county and to the weighted donors, with a placebo standard error of 0.113 and a confidence interval of $[0.051, 0.494]$ that excludes zero. At the **2019** reappraisal the same statistic is -0.080 $[-0.159, -0.002]$: no increase, and if anything a small decline. The point estimates differ in sign and the intervals barely overlap.

Section 5.1 supplies the reason, and it is not the freeze. Both reappraisals settled the same kind of accumulated bill, and the freeze operated identically in both cycles. What differed is the valuation. The roll set in 2011 left the cheapest quartile assessed at 1.11 of sale price against 0.99 for the most expensive: regressive, and the figure is a lower bound because it is observed a year into the freeze. The roll set in 2019 was level across the value distribution, 0.938 against 0.964 once discounted to the January valuation date. Distress followed the regressivity and not the catch-up. A reappraisal that corrects an accumulated gap without misvaluing anyone appears to be absorbed; a reappraisal that also imposes a valuation error on the cheapest homes is not.

I hold that reading loosely, for four reasons the table and the diagnostics make plain, and I would not want it carried further than the evidence supports.

Table 9: Synthetic difference in differences, both reappraisals.

Outcome	Synthetic DiD					DID	SC	Donors
	$\hat{\tau}$	(s.e.)	95% CI	p	p^{RW}	$\hat{\tau}$	$\hat{\tau}$	
<i>2011 reappraisal, $T_{pre} = 16, T_{post} = 16$ quarters</i>								
Foreclosure filing	+0.273	(0.113)	[+0.051, +0.494]	0.04	0.04	+0.123	+0.283	33
Distressed sale	-0.114	(0.041)	[−0.194, −0.034]	0.04	0.04	-0.074	+0.057	33
Transaction	-0.207	(0.164)	[−0.527, +0.114]	0.16	0.16	-0.305	-0.276	33
Log sale price	+0.778	(9.180)	[−17.215, +18.771]	0.93	0.93	-3.551	+0.938	31
Log price per sq. ft.	+0.143	(10.060)	[−19.576, +19.861]	1.00	1.00	-3.755	+0.076	31
<i>2019 reappraisal, $T_{pre} = 16, T_{post} = 8$ quarters</i>								
Foreclosure filing	-0.080	(0.040)	[−0.159, −0.002]	0.09	0.09	-0.089	-0.044	34
Distressed sale	-0.025	(0.013)	[−0.050, −0.001]	0.06	0.06	-0.053	-0.014	34
Transaction	-0.106	(0.123)	[−0.347, +0.135]	0.43	0.43	-0.114	-0.051	34
Log sale price	+5.921	(7.608)	[−8.991, +20.833]	0.27	0.43	+8.282	+2.119	32
Log price per sq. ft.	+4.197	(7.132)	[−9.781, +18.176]	0.32	0.43	+7.213	-1.418	32

Notes. $\hat{\tau}$ from equation (4), estimated on the county-by-quarter panel of quartile gaps defined in (3). Rate outcomes in percentage points, price outcomes in log points. Standard errors are the placebo variance of Arkhangelsky et al. (2021), Algorithm 4, appropriate for a single treated unit; p is the share of placebo draws exceeding the estimate in absolute value and p^{RW} a stepdown adjustment across the ten cells. DID weights all donors and pre-quarters equally; SC is the unregularised synthetic control without an intercept. Donors are the counties satisfying the five conditions of Section 4, all of which reassess annually; the price rows additionally require twenty-five sales in the thinnest quartile-quarter cell.

The crisis sits inside the 2011 window. That window runs 2007 to 2014, and Mecklenburg’s own quartile gap in filings falls steeply through those years as the crisis unwound, from 1.31 percentage points in 2012 to 0.23 in 2017. SDID is built for exactly this problem, and the time weights in Figure 2 show it working: the estimator discards the free-fall quarters and concentrates on the pre-quarters that predict

where the donors were heading. But no weighting scheme recovers a counterfactual the data do not contain, and a positive filing estimate in a crisis window from a single treated county is not the same evidence as the same estimate from a quiet one.

The in-time placebo is not clean. Placing a pseudo-event in the middle of the pre-period, on pre-period data alone, returns +0.099 on filings against +0.273 at the true event. The true estimate is nearly three times the placebo and the placebo’s own interval covers zero, so the test is passed rather than failed. It is not passed comfortably.

Completed distress moves the other way. Distressed sales fall by 0.114 percentage points at 2011, with an interval excluding zero, while filings rise. More filings and fewer completed foreclosure sales is not incoherent, since a filing is the start of a process that a moratorium, a modification or a cure can end, and the 2011–2014 window is thick with all three. But the paper cannot observe which, and two significant coefficients of opposite sign among ten tests warrant caution rather than a mechanism.

The price outcomes are uninformative and are reported as such. The 2011 log price estimate carries a standard error of 9.2 log points and the price per square foot 10.1. The outcome is a difference of two quartile means within a county-quarter, so its variance is the sum of two cell variances, and for donor counties with thin quarterly sale counts the gap is mostly noise even after the minimum-cell screen. This design is powered for the rate outcomes and not for the price outcomes, and the price rows should be read as establishing nothing rather than as null results.

What the donor screen buys. Table 10 sets the eligible pool against an unscreened one admitting any county with a complete series. The unscreened pool is larger and worse: it contains counties on multi-year reappraisal cycles, which reset their own rolls at dates this paper neither controls nor observes, so a fraction of the “controls” are themselves treated. It moves the 2011 filing estimate from 0.273 to 0.428 and the price estimates by eight log points. The screened estimates are the ones to read.

Table 10: Eligible against unscreened donor pools, 2011 reappraisal.

Outcome	Eligible pool		Unscreened	
	Donors	$\hat{\tau}$	Donors	$\hat{\tau}$
Foreclosure filing	33	+0.273	318	+0.428
Distressed sale	33	-0.114	277	-0.118
Transaction	33	-0.207	356	-0.276
Log sale price	31	+0.778	141	-7.215
Log price per sq. ft.	31	+0.143	134	-8.848

Notes. The eligible pool applies the five conditions of Section 4, the first of which requires annual reassessment. The unscreened pool admits any county with a complete, non-degenerate series over the window and therefore includes counties that reappraise on multi-year cycles of their own.

Table 11: Estimator diagnostics, 2011 reappraisal.

Outcome	ζ	Pre-period RMSE	Residual s.d.		In-time placebo
			Meck.	donor med.	
Foreclosure filing	0.380	0.093	0.096	0.100	+0.099
Distressed sale	0.142	0.048	0.049	0.028	-0.048
Transaction	0.409	0.054	0.056	0.120	+0.049
Log sale price	22.735	5.350	5.526	5.902	+0.335
Log price per sq. ft.	21.854	4.598	4.749	5.432	-0.272

Notes. ζ is the regularisation parameter of Arkhangelsky et al. (2021) equation (5). Pre-period RMSE is the discrepancy between Mecklenburg’s demeaned path and its weighted donor path over the pre-window. The residual standard deviations test the homoskedasticity the placebo variance requires: Mecklenburg’s pre-period residual against its synthetic path, beside the median of the same quantity computed for each donor against its own leave-one-out synthetic path. On filings Mecklenburg is quieter than three donors in five, so the placebo standard error is conservative. The in-time placebo places a pseudo-event at the midpoint of the pre-window and estimates on pre-window data alone.

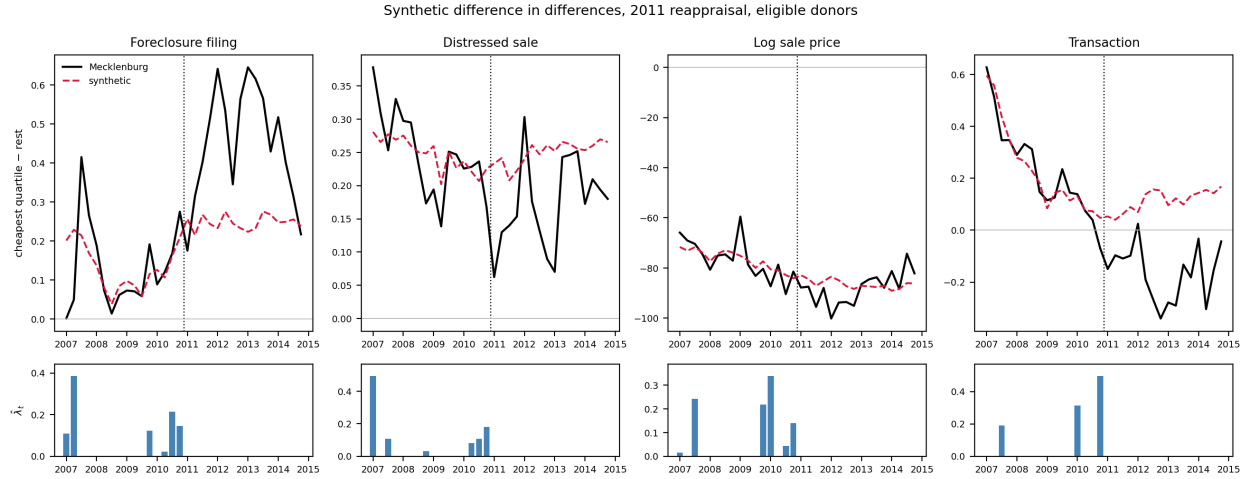


Figure 2: Synthetic difference in differences, 2011 reappraisal.

Notes. Upper panels: Mecklenburg’s quartile gap from (3) against its weighted donor counterpart, the latter shifted by the pre-period difference in means, since SDID matches trends rather than levels and the unit fixed effects absorb the rest. The vertical rule marks the reappraisal. Lower panels: the time weights $\hat{\lambda}_t$ from (6). Their concentration is the estimator’s own judgement about which pre-quarters carry information about the counterfactual, and in a window opening in 2007 that judgement is substantive.

5.4 Weighting on covariates instead of on paths

Synthetic difference in differences chooses donor weights so that the donors’ pre-period outcome *path* runs parallel to Mecklenburg’s. An older and larger literature chooses weights so that the donors’ *covariates* match the treated unit’s, on the argument that treatment is ignorable given observed characteristics (Rosenbaum and Rubin, 1983). The two rest on different assumptions, so if they agree the agreement is corroboration rather than arithmetic. Table 12 runs the comparison.

Four weighting schemes are set beside the main specification, all written in the form $\hat{\tau} = \Delta_0 - \sum_i w_i \Delta_i$

with Δ_i the change in county i 's quartile gap from its pre-period mean to its post-period mean, so they differ only in w . Difference in differences weights every donor equally. Inverse probability weighting uses the odds of treatment from a propensity score (Abadie, 2005), fitted by penalised likelihood (Firth, 1993) because a single treated unit separates an ordinary logit perfectly. Entropy balancing chooses the weights closest to uniform that reproduce Mecklenburg's covariate means exactly (Hainmueller, 2012). The doubly robust estimator combines the propensity weights with an outcome regression and is consistent if either is correctly specified (Sant'Anna and Zhao, 2020). Covariates are pre-period only: the level, trend and volatility of the county's own quartile gap, its owner-occupied parcel count, and the level of the cheapest quartile's outcome.

Table 12: Synthetic difference in differences against covariate-based weighting.

	SDID	DID	IPW	Entropy balance	Doubly robust
<i>Foreclosure filing, 2011 reappraisal</i>					
Estimate	+0.273 (0.113)	+0.123 (0.248)	-0.083 (0.296)	+0.258 (0.211)	+0.325 (0.139)
<i>Foreclosure filing, 2019 reappraisal</i>					
Estimate	-0.080 (0.040)	-0.089 (0.037)	-0.099 (0.043)	-0.070 (0.031)	-0.038 (0.020)
<i>Distressed sale, 2011 reappraisal</i>					
Estimate	-0.114 (0.041)	-0.074 (0.047)	-0.083 (0.052)	-0.049 (0.046)	+0.093 (0.026)
<i>Distressed sale, 2019 reappraisal</i>					
Estimate	-0.025 (0.013)	-0.053 (0.016)	-0.038 (0.017)	-0.047 (0.013)	-0.018 (0.009)

Notes. Percentage points. All five estimators take the form $\hat{\tau} = \Delta_0 - \sum_i w_i \Delta_i$ on the donor pool of Table 9 and differ only in the weights. Standard errors are the placebo variance of Arkhangelsky et al. (2021) Algorithm 4, run through each weighting scheme in turn so the columns are comparable.

Entropy balancing agrees with the main specification. At the 2011 reappraisal it returns +0.258 against the synthetic difference-in-differences estimate of +0.273, and at 2019 -0.070 against -0.080 . Two estimators that select donors on different information, one on the shape of the pre-period path and one on the level of pre-period characteristics, land within two hundredths of a percentage point of each other at both events. I regard that as the most useful thing in this table.

A fitted propensity score does not work here, and the balance table shows why. Table 13 reports standardised differences between Mecklenburg and the weighted donors. Unweighted, the worst covariate is off by 0.93 standard deviations. Inverse probability weighting makes it *worse*, 0.93 to 1.96. That is a diagnostic failure rather than a close call: with one treated unit the propensity score is estimated from a single positive observation, and the resulting weights concentrate on donors that happen to sit where the fitted score is steepest rather than on donors that resemble the treated county. The -0.083 it returns for

2011 filings, the only estimate in the table with the wrong sign, should be read as an artefact of that failure. The doubly robust estimator inherits the same weights, and although it performs well on filings it returns +0.093 for distressed sales at 2011 where every other estimator is negative.

Entropy balancing avoids this by construction: it imposes balance rather than fitting a model and hoping for it, which is the argument Hainmueller (2012) makes for the method and which applies with unusual force when the treated group has one member.

Table 13: Covariate balance under each weighting scheme, 2011.

Covariate	Unweighted	IPW	Entropy balance	SDID
Pre-period gap, level	+0.893	+1.960	-0.000	+0.474
Pre-period gap, trend	+0.439	+1.275	-0.000	+0.230
Pre-period gap, s.d.	-0.553	-1.462	-0.000	-0.230
Log parcels	+0.930	+0.410	+0.000	+1.102
Cheapest quartile, level	-0.164	-0.703	-0.000	+0.093
Worst absolute difference	0.930	1.960	0.000	1.102

Notes. Standardised differences between Mecklenburg and the weighted donors, scaled by the unweighted donor standard deviation, on the pre-period covariates used to fit the weights. Values near zero indicate balance; 0.25 is the threshold Section 4 applies to its anchor screen. Entropy balancing achieves exact balance by construction. The synthetic difference-in-differences column is reported for completeness, not as a criterion the estimator is trying to meet.

The synthetic control weights do not balance covariates, and are not meant to. Their worst standardised difference is 1.10, driven by county size: the donors whose quartile gaps track Mecklenburg’s are not the donors that look like Mecklenburg on paper. That is a feature of the estimator rather than a defect, since parallel trends in the outcome is what the design requires and covariate similarity is neither necessary nor sufficient for it. But it is worth stating, because a reader who assumes the synthetic county resembles Mecklenburg in its observables will be surprised, and because it explains why the two routes are genuine alternatives rather than the same procedure in different clothing.

5.5 How much of the excess distress does the reappraisal account for?

The estimates in Table 9 are effects. A reader will want the magnitude expressed against something, and the natural benchmark is the excess the cheapest quartile carries in the first place. Mecklenburg’s cheapest homes file for foreclosure more often than the rest of the county at both reappraisals and in every quarter between them, and only part of that excess is anything a reappraisal did.

Write the observed excess over the post-period as

$$G = \frac{1}{|\mathcal{T}^+|} \sum_{t \in \mathcal{T}^+} \left(\bar{y}_{0t}^{\text{Q1}} - \frac{1}{3} \sum_{q=2}^4 \bar{y}_{0t}^q \right), \quad (7)$$

the same contrast (3) forms, averaged over the post-period for the treated county alone. G is a description of Mecklenburg and involves no counterfactual. The synthetic difference-in-differences estimate is the part of it the reappraisal accounts for, so

$$\underbrace{G}_{\text{observed excess}} = \underbrace{\hat{\tau}}_{\text{the reappraisal}} + \underbrace{(G - \hat{\tau})}_{\text{standing gap}}, \quad \hat{s} = \hat{\tau}/G. \quad (8)$$

Because G is observed exactly, all the uncertainty in $\hat{s}$ is uncertainty about $\hat{\tau}$, and the interval for the share is the interval for the effect divided by G . Two hypotheses are worth testing. That $s = 0$ says the reappraisal accounts for none of the excess. That $s = 1$ says it accounts for all of it, which would mean the cheapest quartile carried no standing disadvantage of its own.

Table 14: What share of the cheapest quartile’s excess filing rate the reappraisal accounts for.

Event	Observed, per quarter (%)			Decomposition		Share		p $H_0: s = 1$
	Q1	Q2–Q4	Excess G	$\hat{\tau}$	$G - \hat{\tau}$	$\hat{s}$	95% CI	
2011	1.160	0.709	0.450	+0.273	0.177	61%	[11, 110]	0.116
2019	0.053	0.041	0.013	-0.080	0.093	—	—	—

Notes. Quarterly foreclosure filing rates in percent, averaged over each event’s post-period. G is the observed excess of the cheapest quartile over the mean of the other three, from equation (7); $\hat{\tau}$ is the synthetic difference-in-differences estimate from Table 9; $\hat{s} = \hat{\tau}/G$ and its interval is the interval for $\hat{\tau}$ divided by G . The share is suppressed at 2019 because the observed excess there, 0.013 points, is smaller than the standard error of the estimate, so it is a ratio to a denominator indistinguishable from zero.

At the 2011 reappraisal the cheapest quartile filed at 1.160 percent per quarter against 0.709 for the rest of the county, an excess of 0.450 points. The estimate attributes 0.273 of that to the reappraisal, so $\hat{s} = 61$ percent with an interval of 11 to 110. Two readings follow and they should be kept apart.

The lower end is what the design establishes. The hypothesis that the reappraisal accounts for none of the excess is rejected, $p = 0.016$, and at least a tenth of it is attributable at conventional confidence. Over sixteen post-period quarters and 33,903 bottom-quartile parcels, the point estimate corresponds to roughly 1,480 foreclosure filings that the reappraisal accounts for.

The upper end is what the design does *not* establish. The interval reaches 110 percent and the hypothesis that $s = 1$ is not rejected, $p = 0.116$. The data are therefore consistent with the reappraisal accounting for the entire excess, and I cannot claim from this evidence that the cheapest quartile’s disadvantage is partly

structural. The standing gap of 0.177 points is a point estimate with an interval that covers zero.

Decomposing the two sides. Reading $\hat{\tau}$ as an effect on the cheapest quartile’s own filing rate, rather than on its excess, requires that the reappraisal left quartiles two to four alone. In the contrast of (3) those quartiles are the comparison group, so the assumption cannot be tested there. Estimating each side against its own synthetic control makes it testable. Table 15 fits the cheapest quartile’s filing rate and the mean of the other three separately, on the same donor pool and windows. The placebo loop draws the same donor for both series at each replication, so the two estimates are paired and the variance of their difference is estimated rather than assumed.

Table 15: Each side of the contrast against its own synthetic control.

Event	Cheapest quartile		Quartiles two to four			Gap, per quarter (%)			
	$\hat{\tau}_{Q1}$	(s.e.)	$\hat{\tau}_{rest}$	(s.e.)	p	Observed	Standing	(s.e.)	p
2011	+0.547	(0.255)	+0.217	(0.178)	0.222	0.450	0.120	(0.126)	0.339
2019	-0.081	(0.058)	-0.025	(0.047)	0.586	0.013	0.068	(0.043)	0.116

Notes. Quarterly foreclosure filing rates in percent. $\hat{\tau}_{Q1}$ and $\hat{\tau}_{rest}$ are separate synthetic difference-in-differences fits, each with its own unit weights, time weights and placebo variance, on the donor pool of Table 9. The standing gap is the difference of the two counterfactual levels, observed minus estimated effect on each side; its standard error comes from paired placebo draws, which share a donor at each replication and so carry the covariance between the two estimates. That correlation is 0.89 at 2011 and 0.67 at 2019, which is why the difference is estimated more precisely than either level.

The test passes at both events. The reappraisal’s estimated effect on quartiles two to four is +0.217 at 2011 and −0.025 at 2019, neither distinguishable from zero, $p = 0.222$ and $p = 0.586$. This is a failure to reject rather than a demonstration: +0.217 is forty percent of the effect on the cheapest quartile and is imprecise rather than small. But nothing in the data contradicts reading the contrast as an effect concentrated on the cheapest homes.

The decomposition corrects one figure and sharpens another. The effect on the cheapest quartile’s own filing rate is +0.547 points, not the +0.273 the gap estimate carries, because the reappraisal moved the comparison group in the same direction and the gap nets that movement out. Against an observed rate of 1.160 the reappraisal therefore accounts for closer to a half of the cheapest quartile’s filing rate than to the quarter implied by the gap estimate. And the implied effect on the gap, +0.330 with a standard error of 0.126, reproduces the +0.273 of Table 9 from a different set of weights, which is the consistency check the decomposition exists to provide.

The standing gap is 0.120 points with a standard error of 0.126, so the counterfactual excess is not distinguishable from zero, $p = 0.339$. That agrees with the test of $s = 1$ above and settles the question the same way: the data do not establish that the cheapest quartile carried a disadvantage of its own absent the reappraisal, and they do not exclude one either.

What the decomposition costs. The gap in (3) differences out anything that moved every quartile

in the county together, which is the whole reason the contrast is robust in a window containing the financial crisis. The two level components give that protection up: each is identified only if the donor pool absorbs the county-wide shock, which is a stronger requirement than the gap design needs. The correlation of 0.89 between the paired placebo draws is direct evidence of how much the two series share. I therefore keep the gap estimate as the headline and use the decomposition for what it is good for, testing the comparison-group assumption and putting the effect in levels.

5.6 From filings to households

A filing is the start of a process. Turning the estimated effect on filings into households that lost a home takes three quantities, and Table 16 estimates each of them. The first is a level effect by quartile. The gap estimate of Table 9 cannot be split across quartiles, so each quartile’s own filing series is fitted against its own synthetic control, on the donor pool and window of that table. The second is the share of filings that reach a completed distressed transfer, measured in the Mecklenburg panel within three years of the filing and computed separately by quartile, since cure rates move with equity. The third is the count of parcels at risk.

Multiplying through, the 2011 reappraisal accounts for 423 bottom-quartile households losing a home, 9.9 for every thousand homes in the quartile, with a 95 percent interval of 23 to 823. The other three quartiles carry point estimates of 345, 180 and 78, and none of the three effects they rest on is distinguishable from zero. The completion rate is what separates the household count from the filing count: 11.3 percent of bottom-quartile filings reach a completed transfer within three years, so the great majority end in a cure.

Table 16: Households displaced at the 2011 reappraisal.

Quartile	Filing effect		Parcels	Completion rate	Attributable filings	Households displaced	Per 1,000 homes
	pp/qtr	(s.e.)					
Q1*	+0.547	(0.264)	42,682	11.3%	3,736	423	9.9
Q2	+0.402	(0.242)	49,393	10.8%	3,181	345	7.0
Q3	+0.190	(0.205)	51,290	11.6%	1,560	180	3.5
Q4	+0.069	(0.155)	51,021	13.9%	559	78	1.5

Notes. The filing effect is a separate synthetic difference-in-differences fit for each quartile’s own quarterly filing rate, with its own unit weights, time weights and placebo variance, on the donor pool and window of Table 9; a level effect requires the pool to absorb the county-wide shock, which is the stronger assumption discussed in Section 5.5. The completion rate is the share of filings on owner-occupied parcels reaching a completed distressed transfer within three years, measured in the Mecklenburg panel. Attributable filings are the effect per quarter times sixteen post-period quarters times the parcel count; households displaced applies the completion rate to that product. An asterisk marks a filing effect distinguishable from zero at 95 percent.

Where those households went. A count says how many. To say what happened to them I match parcels to a consumer address history that records up to ten addresses for each person with the month each took effect, described in Section 4. Matching on postal code, house number and street name links 96.8 percent

of Mecklenburg parcels, and the rate is flat across the value distribution, 99.1 percent in the cheapest quartile against 98.7 in the most expensive, so the link does not select the homes this exercise is about. Ordering a person's addresses by their effective dates makes the address they moved to next observable, so a departure from a Mecklenburg parcel carries a destination.

The link registers the departures it is supposed to register. In every quartile a parcel that received a filing shows a later departure more often than a parcel that did not, and Table 19 gives the size of that difference.

Table 17 sets the destinations of the two groups side by side. Households leaving a bottom-quartile home after a filing left the county less often than households leaving one without, and moved to an address carrying a unit number more often. Among those staying in the county, they moved to a cheaper block group relative to the one they left. Both groups moved to more expensive neighbourhoods on average, which is what the cheapest quartile having little below it implies; the difference between them is what carries the content.

That difference appears at every quartile and at a similar size, -0.090 log points in the cheapest and -0.128 , -0.157 and -0.120 in the others, each with a standard error under 0.03. A filing is followed by a move to a cheaper neighbourhood than the household's own quartile peers make, wherever in the value distribution it starts.

Applying the cheapest quartile's shares to the 423 households of Table 16 puts roughly 142 of them outside Mecklenburg County, 97 outside North Carolina, and 95 at an address with a unit number. Those figures inherit the interval on the count, which is wide.

Table 17: Where departures from Mecklenburg parcels went, 2011 to 2016.

	Departures	Left county	Left state	To an apartment	Within the county		
					To a cheaper block group	Change in log value	Change in quartile
<i>Q1 (cheapest)</i>							
filing	1,986	33.6%	22.9%	22.6%	18.6%	+0.203	+0.80
no filing	28,641	42.7%	26.2%	19.5%	13.0%	+0.294	+0.97
<i>Q2</i>							
filing	2,448	41.7%	26.8%	20.7%	42.8%	+0.021	+0.11
no filing	41,422	50.8%	30.9%	17.1%	29.1%	+0.149	+0.45
<i>Q3</i>							
filing	1,598	45.4%	28.7%	19.2%	51.7%	-0.078	-0.41
no filing	49,313	54.0%	33.5%	15.3%	37.2%	+0.079	+0.04
<i>Q4 (most expensive)</i>							
filing	861	42.4%	28.5%	16.6%	63.7%	-0.270	-0.57
no filing	51,561	49.2%	32.8%	15.4%	56.2%	-0.150	-0.45

Notes. One row per person leaving an address at a Mecklenburg parcel between 2011 and 2016, split by whether that parcel received a foreclosure filing between 2011 and 2014. Destination is the next address in the person's history. An apartment is an address carrying a unit number. The last three columns are computed on departures whose destination is in Mecklenburg County; the block-group price columns additionally require the destination address to match a parcel the estimation sample carries a price for, which holds for 43.7 percent of in-county destinations. Table A6 describes the link in full. Change in log value is the log median block-group sale price of the destination minus that of the origin, on the sort of Section 4; change in quartile is the same comparison in quartile units and is bounded from below in the cheapest quartile.

Bought or rented. A unit number is a proxy for renting, and it misses the single-family rental, which in a Sunbelt county after 2009 is where a foreclosed owner-occupant is most likely to land. Table 18 replaces the proxy with a deed check. A destination inside the county that matches a parcel is looked up in the county's recorded arm's-length transfers: a household with a transfer at that parcel within a year of arriving bought the home, and a household with none moved into a home somebody else owns. The check is defined on movers who stay in the county, since no deed record outside Mecklenburg is available, so the share leaving the county is carried in the table beside it.

Among cheapest-quartile households leaving after a filing and staying in the county, 14.2 percent bought the next home. Of the rest, 44.0 percent moved into a single-family home they did not buy and 41.8 percent into multifamily housing, so 85.8 percent became renters, against 78.7 percent of their quartile peers leaving without a filing: a difference of 7.2 points with a standard error of 1.0. Read as a purchase probability the same contrast is sharper. A filing costs a household between a third and a half of its chance of buying again, and it does so at every quartile: the loss is 7.2 points against a base of 21.3 percent in the cheapest and 16.9 against 36.9 in the third, where the ordinary mover is far more likely to be buying in the first place.

Applying the cheapest quartile's shares to the 423 households of Table 16, 281 stay in the county, of whom about 40 buy, 124 rent a house and 118 move into multifamily housing. The 142 who leave the county carry no tenure measure.

Table 18: Did the household buy the next home? Destination tenure from the deed record.

	In-county	Bought	Single-family		Renter share	
			rental	Multifamily	± 1 yr	± 2 yr
<i>Q1 (cheapest)</i>						
filing	1,319	14.2%	44.0%	41.8%	85.8%	81.3%
no filing	16,399	21.3%	41.5%	37.2%	78.7%	74.8%
<i>Q2</i>						
filing	1,428	18.3%	41.9%	39.7%	81.7%	76.5%
no filing	20,365	27.8%	37.8%	34.4%	72.2%	67.6%
<i>Q3</i>						
filing	872	20.1%	44.5%	35.4%	79.9%	72.5%
no filing	22,698	36.9%	32.6%	30.5%	63.1%	58.3%
<i>Q4 (most expensive)</i>						
filing	496	27.4%	39.5%	33.1%	72.6%	65.7%
no filing	26,199	42.1%	28.7%	29.2%	57.9%	53.2%

Notes. Departures of Table 17 whose destination is inside Mecklenburg County, classified by the county's recorded transfers. Bought is a destination matching a parcel in the assessor file at which an arm's-length transfer is recorded within one year either side of the month the household arrived; single-family rental is a matching parcel with no such transfer; multifamily is an address carrying a unit number or matching no parcel. The renter share is the second and third together, reported at a one-year and a two-year transfer window. Mortgage records are excluded by the arm's-length flag, which matters in this county because they outnumber deeds and carry the loan amount in the price field. Three conventions bound the check. An owner-occupied condominium carries a unit number and is counted as multifamily, so the purchase share is a lower bound; running the deed check on those addresses as well moves 0.5 percent of in-county moves. Widening the window to two years raises the overall purchase share from 32.7 to 37.3 percent and leaves the gradient across filing status intact. And a household moving into a home a relative owns records no transfer and is counted as renting. As a check that the test is measuring tenure rather than gaps in the deed record, the assessor's own occupancy flag marks 85.8 percent of the purchase class's destination parcels owner-occupied against 70.2 percent of the single-family rental class's; that flag is a recent snapshot, so the gap is directional.

Reading the contrast across quartiles. Table 19 collects every measure this section draws from the address history into one comparison, filing against no filing, quartile by quartile, so each difference and its standard error sit beside the two levels that produce them. Three readings follow, and the middle of the value distribution qualifies two of them.

Households at parcels that received a foreclosure filing are 14.5 to 20.6 percentage points more likely to leave in the four years after the reappraisal. The gap is largest in the cheapest quartile, 20.6 points against 17.7 in the most expensive, a factor of 1.2. The loss of a home after the reappraisal therefore falls more often on cheaper homes. The ordering in between does not follow the value distribution: the third quartile's gap is the smallest of the four, at 14.5 points.

Among households that do move, those that received a filing are 6.8 to 9.2 points more likely to stay in the county. That gap does follow the value distribution, running 9.2 points in each of the two cheapest quartiles against 6.8 in the most expensive, a factor of 1.4. A filing appears to narrow the range of places a household can go, and the narrowing is concentrated among cheaper homes. The comparable gap for staying

in the state runs 3.3 to 4.8 points and is smallest at the bottom, so the county margin is the one carrying the distributional claim.

Among households that do move, those that received a filing are 1.2 to 3.9 points more likely to leave a single-family home for an apartment. The gap runs 3.1 points in the cheapest quartile against 1.2 in the most expensive, a factor of 2.5. Households that lose a home take lower-quality housing in exchange. Here too the middle of the distribution breaks the ordering, the third quartile's gap being the largest of the four at 3.9 points.

Table 19: Departures from a parcel with a foreclosure filing against departures from one without, 2011 event.

	Filing	No filing	Difference
<i>Parcels with an observed departure</i>			
Q1	57.5%	37.0%	+20.6pp (1.2)
Q2	59.0%	42.2%	+16.7pp (1.2)
Q3	58.9%	44.4%	+14.5pp (1.5)
Q4	61.6%	43.9%	+17.7pp (2.0)
<i>Left Mecklenburg County</i>			
Q1	33.6%	42.7%	−9.2pp (1.1)
Q2	41.7%	50.8%	−9.2pp (1.0)
Q3	45.4%	54.0%	−8.5pp (1.3)
Q4	42.4%	49.2%	−6.8pp (1.7)
<i>Left North Carolina</i>			
Q1	22.9%	26.2%	−3.3pp (1.0)
Q2	26.8%	30.9%	−4.1pp (0.9)
Q3	28.7%	33.5%	−4.8pp (1.2)
Q4	28.5%	32.8%	−4.3pp (1.6)
<i>Destination carries a unit number</i>			
Q1	22.6%	19.5%	+3.1pp (1.0)
Q2	20.7%	17.1%	+3.7pp (0.8)
Q3	19.2%	15.3%	+3.9pp (1.0)
Q4	16.6%	15.4%	+1.2pp (1.3)
<i>Moved to a cheaper block group</i>			
Q1	18.6%	13.0%	+5.6pp (1.9)
Q2	42.8%	29.1%	+13.7pp (2.3)
Q3	51.7%	37.2%	+14.5pp (2.7)
Q4	63.7%	56.2%	+7.5pp (3.4)
<i>Change in log block-group value</i>			
Q1	+0.203	+0.294	−0.090 (0.016)
Q2	+0.021	+0.149	−0.128 (0.015)
Q3	−0.078	+0.079	−0.157 (0.016)
Q4	−0.270	−0.150	−0.120 (0.027)
<i>Change in value quartile</i>			
Q1	+0.800	+0.971	−0.170 (0.044)
Q2	+0.110	+0.455	−0.345 (0.042)
Q3	−0.407	+0.041	−0.448 (0.050)
Q4	−0.567	−0.446	−0.121 (0.054)
<i>Bought the next home</i>			
Q1	14.2%	21.3%	−7.2pp (1.0)
Q2	18.3%	27.8%	−9.5pp (1.1)
Q3	20.1%	36.9%	−16.9pp (1.4)
Q4	27.4%	42.1%	−14.7pp (2.0)
<i>Rented a single-family home</i>			
Q1	44.0%	41.5%	+2.5pp (1.4)
Q2	41.9%	37.8%	+4.1pp (1.3)
Q3	44.5%	32.6%	+11.9pp (1.7)
Q4	39.5%	28.7%	+10.8pp (2.2)
<i>Moved into multifamily housing</i>			
Q1	41.8%	37.2%	+4.7pp (1.4)
Q2	39.7%	34.4%	+5.3pp (1.3)
Q3	35.4%	30.5%	+5.0pp (1.6)
Q4	33.1%	29.2%	+3.9pp (2.1)
<i>Became a renter</i>			
Q1	85.8%	78.7%	+7.2pp (1.0)
Q2	81.7%	72.2%	+9.5pp (1.1)
Q3	79.9%	63.1%	+16.9pp (1.4)
Q4	72.6%	57.9%	+14.7pp (2.0)

Notes. Filings are those recorded between 2011 and 2014 and departures those dated 2011 to 2016, the sample of Table 17. The first block is defined on parcels and the next three on departures; the remaining seven are defined on departures whose destination is inside Mecklenburg County, since neither a block-group price nor a deed record is available elsewhere. Shares carry a binomial standard error and the two changes the standard error of the mean; each difference carries the root of the two summed. Destination tenure is the deed check of Table 18 at its one-year window. Standard errors are in the units of the difference beside them, percentage points or log points.

What this comparison does not establish. Three limits bound it. The address history is a current-vintage file, so a person appears in it because they are traceable today, and a household that dissolved is absent; whether a departure is observed at all is therefore an outcome in its own right, and the departure rates above are reported for that reason. The link places a person at an address rather than on a deed, so a tenant or a later occupant can match a parcel. And nothing here is estimated against a counterfactual. The count in Table 16 is the causal object, and this is a description of what befell households of that kind at that time. Every difference in Table 19 sets households that lost a home beside households that moved for other reasons inside the same county, so it nets out nothing that moved Mecklenburg as a whole between 2011 and 2016. Running the same comparison in the donor counties and differencing would, and that is the natural extension of this exercise.

5.7 Does the size of the bill matter?

Every estimate so far compares groups of homes. None asks whether the households whose bills rose most are the households whose distress rose most, and that is the question a liquidity mechanism actually makes a prediction about. The administrative levy panel answers it, because the correction a parcel received at the reappraisal varies widely within its own quartile.

I regress the foreclosure-filing indicator on the parcel’s own percent change in the tax bill, with calendar-year and quartile effects and the controls of Section 3.1. Identification comes from two parcels in the same quartile and the same year receiving different corrections, so nothing common to the county, to the year or to the value group can drive the coefficient.

Table 20: Foreclosure filings on the parcel’s own bill change.

Sample	Effect per point of bill change (pp)			Parcel-years
	Coefficient	(s.e.)	<i>t</i>	
All quartiles	-0.00206	(0.00078)	-2.65	597,995
Q1	+0.00102	(0.00158)	+0.64	134,107
Q2	+0.00126	(0.00224)	+0.56	157,674
Q3	+0.00106	(0.00092)	+1.15	136,707
Q4	-0.00112	(0.00072)	-1.55	169,507

Notes. Owner-occupied Mecklenburg parcels, 2013–2020. The dependent variable is the foreclosure-filing indicator and the regressor is the parcel’s own percent change in the administrative tax bill, winsorised at the 1st and 99th percentiles as elsewhere. Calendar-year effects throughout, quartile effects in the pooled row, and the tract house-price index and structure characteristics as controls. Standard errors clustered by census tract. The within-quartile standard deviation of the 2019 change is 16.6 points, so the dose varies substantially inside each value group.

The answer is no. Within every quartile the coefficient is indistinguishable from zero, and the pooled

estimate is small, negative and statistically significant, which is the wrong sign for a liquidity mechanism. Scaled to the cheapest quartile’s own 28.2-point increase, the pooled coefficient implies -0.058 percentage points.

This is the sharpest evidence in the paper against the mechanism the literature would expect, and it is worth being clear about what it does and does not rule out. It does not say a reappraisal cannot cause distress; the 2011 estimate in Section 5.3 says otherwise. It says that among households facing the same reappraisal in the same year and the same part of the value distribution, the ones receiving larger corrections did not file more often. A story in which a bigger bill causes more distress predicts a positive within-quartile slope, and there is none. Whatever distinguished the cheapest quartile at the 2011 reappraisal, it was not the size of the individual increase that household received.

That result sits more comfortably with the magnitudes of Section 5.9 than a positive slope would. The cheapest quartile’s correction was \$33 a month against a home that had appreciated 61 log points. Variation around a shock that small, in a group whose equity was rising throughout, is not where one should expect to find a foreclosure margin.

5.8 Selection into sale

The price outcome is observed only on parcel-years carrying a sale, and the reappraisal moves the probability of a sale in every quartile. The composition of sellers therefore changes with treatment, and under a liquidity mechanism it changes toward distressed sellers, which is the direction of the hypothesised effect. Table 21 reports trimming bounds after Lee (2009), which require only that the reappraisal move selection in the same direction for every parcel. The bounds contain zero in every quartile and for the cheapest-against-most-expensive contrast, where they run from -17.1 to $+18.3$ percent. Whatever a price regression on this sample reports, its sign is not identified once selection into sale is admitted.

Table 21: Trimming bounds on the price effect.

	Trimmed (%)	Naive (%)	Lee bounds (%)	95% CI on the bounds
Q1 (cheapest)	8.9	-7.96	$[-16.66, +0.02]$	$[-22.53, +4.67]$
Q2	10.0	-5.87	$[-12.42, +0.21]$	$[-15.65, +3.63]$
Q3	15.7	-6.13	$[-16.56, +2.48]$	$[-20.25, +4.86]$
Q4 (most expensive)	9.6	-7.08	$[-18.76, +1.27]$	$[-24.01, +6.37]$
Q1 – Q4	—	-0.88	$[-17.12, +18.30]$	$[-25.81, +24.73]$

Notes. Bounds after Lee (2009) on the effect of the reappraisal on the log sale price, 2019 against 2020, computed within quartile. The trimmed share is the excess selection into sale in the treated year as a fraction of that year’s selection rate; the bounds trim that share from each tail of the treated outcome distribution. Confidence intervals are a tract-clustered bootstrap of the whole procedure, selection rates and trimming included.

5.9 Magnitudes

Table 22 converts the shock into the units a distress mechanism needs. The cheapest quartile’s bill rises \$391 a year, \$33 a month, against a mean 2019 sale price of \$298,000. Capitalised in perpetuity at five percent the tax change is worth 2.6 percent of value, and at three percent 4.4 percent. Full capitalisation is an upper bound, and a generous one here. The roll is reset again within four to eight years and the millage rate is reset annually, so the perpetuity overstates the present value of the shock. The county also cut the combined rate from 1.251 to 0.974 per hundred dollars of assessed value at this reappraisal, so the reset redistributes across the value distribution more than it raises the county’s levy.

This bound is what disciplines the magnitudes a design can plausibly report. An estimated price decline of fifteen percent for the cheapest quartile would be six times what full capitalisation of its own tax change permits. Read together with Section 3.2, the reading is not that the tax has an outsized effect but that such a specification is recovering something other than the tax.

Table 22: The shock in dollars, against full capitalisation.

	Mean bill (\$)		Change		Mean sale price (\$)	Implied $\Delta P/P$ (%)		
	2018	2019	per yr	per mo		$r=3\%$	$r=5\%$	$r=7\%$
Q1 (cheapest)	1,339	1,730	391	33	298,033	-4.38	-2.63	-1.88
Q2	1,773	2,082	309	26	306,796	-3.36	-2.01	-1.44
Q3	2,646	2,925	279	23	411,721	-2.26	-1.35	-0.97
Q4 (most expensive)	5,278	5,433	155	13	718,760	-0.72	-0.43	-0.31

Notes. Mean administrative tax bill by quartile in the year before and the year of the reappraisal, the change, and the change per month; mean sale price on transacting parcel-years in 2019, in 2025 dollars. The implied proportional price effect is $\Delta P/P = -\Delta T/(rP)$, the present value of a permanent change in the annual bill discounted at r , expressed as a share of value. It is an upper bound on capitalisation, not a prediction.

5.10 Where the outcome data end

Table 23 reports the coverage of the treated county’s own series and explains why the estimation window closes in 2020. Recorded foreclosure filings are identically zero in 2022 and 2023, across roughly 110,000 owner-occupied parcels a year, and the sale rate falls to 0.72 and 0.59 percent against 4.33 percent in 2019. Measured against the county’s own best-covered four-year stretch, the same yardstick Section 4 applies to candidate donor counties, those years stand at 0.16 and 0.13, far below the 0.60 a donor must clear. The parcel count is unchanged in those years, so this is the extract ending rather than the market. The levy panel is separately incomplete in 2013 through 2015, holding as little as 21 percent of the county’s parcels in 2014.

A design that used 2020 through 2023 as the counterfactual for the 2019 reappraisal would therefore be comparing a live year against years in which the dependent variable is pinned at zero by the end of the

data. That comparison on its own produces a positive reappraisal coefficient on filings and a negative one on prices.

Table 23: Coverage of the treated county’s own series, by year.

Year	Levy panel		Recorded sales			Foreclosure filings	
	Parcel-yrs	Share	N	Rate (%)	Stability	N	Rate (%)
2012	188,059	0.87	4,750	2.53	0.55	3,146	1.673
2013	117,362	0.54	4,042	3.44	0.75	2,032	1.731
2014	45,943	0.21	1,700	3.70	0.81	567	1.234
2015	108,488	0.50	3,307	3.05	0.66	680	0.627
2016	189,145	0.88	7,660	4.05	0.88	854	0.452
2017	196,720	0.91	10,578	5.38	1.17	594	0.302
2018	198,316	0.92	9,129	4.60	1.00	396	0.200
2019	197,638	0.92	8,553	4.33	0.94	286	0.145
2020	202,505	0.94	7,760	3.83	0.83	138	0.068
2021	213,545	0.99	5,224	2.45	0.53	124	0.058
2022	204,132	0.95	1,460	0.72	0.16	0	0.000
2023	215,495	1.00	1,273	0.59	0.13	0	0.000

Notes. Owner-occupied Mecklenburg parcels. Panel share is parcel-years against the modal year, a property of the assessor levy file. The sale rate is recorded sales per parcel-year, so a thin panel year is not counted twice; stability is that rate against the county’s own best-covered four consecutive years, the test Section 4 applies to donors. The rule marks the close of the estimation window.

6 Conclusion

A property tax that revalues once every eight years is not a tax on current value. It is a tax on value as of the last reset, and between resets it quietly redistributes: homes that appreciate faster than average pay less than a current-value tax would ask, homes that appreciate slower pay more, and the reappraisal collects the difference in one step. Who bears that deferred burden is an empirical question about where prices moved, not a question about how well the assessor works.

In Mecklenburg between 2012 and 2018 prices moved fastest at the bottom. The cheapest quartile of neighbourhoods appreciated 61 log points against 31 for the most expensive, whether measured on repeat sales or on the drift of the frozen assessment ratio across the whole stock. By 2018 those homes were assessed at 0.60 of sale price against 0.73 for the most expensive: under-assessed relative to expensive homes, the reverse of the pattern the regressivity literature describes. The 2019 reappraisal then raised their bills 28.2 percent against 6.8, a spread of 21.5 percentage points where every other year in the panel shows five hundredths of a point.

That gradient is a catch-up and not an imposed error, and two measurements say so. At the reset the assessor is level across the value distribution, the assessed-to-sale ratio running 0.938 in the cheapest quartile against 0.964 in the most expensive once sale prices are discounted to the January valuation date. And the reset closes only 42 percent of the gap the freeze opened, so the cheapest homes leave it still under-

assessed relative to the most expensive. The county's previous roll, measured the same way, was regressive. Regressivity is a property of a valuation; the freeze is a property of the regime, and it operated in both cycles.

I find no differential distress. Calendar-year fixed effects absorb what the county experiences in common. So specified, foreclosure filings among the cheapest quartile move -0.000 percentage points at the reappraisal relative to the most expensive, and their sale prices -0.12 percent. Neither is distinguishable from zero, both sit last in their own placebo distributions, and trimming bounds that admit selection into sale leave the price contrast unsigned. The null is what the mechanism predicts rather than a puzzle. The bill rose because the house was worth more. A bottom-quartile owner pays about \$33 a month more against a home that appreciated 61 log points, and full capitalisation of a tax change that size permits a price movement of 2.6 percent, so any design reporting fifteen percent is recovering something other than the tax.

Three limits bound the claim. Block-group medians measure a neighbourhood rather than a house, so exposure is measured with error and the outcome estimates are attenuated. The design observes one reappraisal cycle, and a single catch-up is not the same object as the cumulative burden of a regime; the pre-period gradients show the financial crisis unwinding differentially across quartiles, which is exactly the kind of confound a longer panel would need to handle. And the incidence documented here is specific to a gentrifying Sunbelt county in one decade. In a county where the top of the market appreciated fastest, the same freeze would defer the same burden onto expensive homes, and the mechanism predicts it.

The policy reading follows from the mechanism rather than the null. If a long cycle taxes on stale values and settles up in one lumpy bill, the lever is the length of the cycle, and its effect is mechanical: halving the interval halves the accumulated divergence and the size of the correction, without requiring any improvement in appraisal. North Carolina applied that lever here, moving Mecklenburg to a four-year cycle after 2019. Whether shortening the cycle changes outcomes for owners, rather than merely smoothing their bills, is the question a second cycle of data would answer.

A Appendix

A.1 Whether a better price index buys a more precise estimate

The price rows of Table 9 establish nothing, and Section 5.3 attributes that to the outcome being a difference of four county-quarter-quartile mean sale prices. A mean sale price moves when the mix of transacting homes moves, which is the problem the house-price index literature was built to solve, so the obvious question is whether one of its answers rescues the estimate. Table A1 reports four, each estimated within county and value quartile and then differenced as before: repeat sales after Bailey et al. (1963), the interval weighting of Case and Shiller (1989), the hybrid of Case and Quigley (1991) that stacks single sales carrying hedonic characteristics onto the resale pairs, and a penalty on the second differences of the index, which is the state-space idea in a form that requires no filter.

None of them helps. Every index carries a larger standard error and a worse pre-period fit than the quality-adjusted quartile mean the paper uses. The ordering among them is exactly what the literature predicts, each refinement improving on the last, but they improve from a much noisier starting point: within a county-quartile-quarter there are hundreds of sales and only a handful of resale *pairs*, so differencing away the house removes composition at the cost of discarding most of the data. The sign turns negative under all four, where the mean-based estimates run positive, which is a further sign that nothing in this outcome is pinned down.

The exercise is reported because it bounds what is achievable rather than because it changes an estimate. Mecklenburg’s quartile price gap moves by seven to nine log points a quarter before the reappraisal under every construction tried, and against sixteen post-quarters and one treated county that volatility floors the standard error near five points. The reappraisal’s own arithmetic, in Table 22, permits a price effect of at most a few points. The design cannot resolve an effect that small, and no measurement of price will make it do so. What the tighter interval does establish is that the large price discounts reported in earlier drafts of this paper are outside it.

Table A1: Price index constructions, 2011 reappraisal.

Price measure	Donors	$\hat{\tau}$	(s.e.)	(stud. s.e.)	Pre-fit	95% CI
Repeat sales	33	-6.00	(16.40)	(7.57)	5.94	[-8.6, +18.7]
Repeat sales, interval weighted	33	-6.87	(14.67)	(7.28)	5.92	[-8.9, +17.5]
Repeat sales and single sales	33	-4.00	(10.06)	(7.49)	5.44	[-8.9, +17.7]
Repeat sales, smoothed	33	-3.62	(8.35)	(7.59)	5.17	[-11.4, +14.6]
<i>Quartile mean, quality adjusted</i>	26	+3.03	(6.11)	(6.63)	4.21	[-20.3, +12.6]

Notes. Log points. Each row re-estimates $\hat{\tau}$ from equation (4) on the 2011 window, changing only how the quarterly price of a county's value quartile is measured. Indices are estimated over 2004 to 2015 within county and quartile and normalised at the first quarter; the event window is read from them. Repeat-sales rows use consecutive same-parcel sale pairs with both sales above ten thousand dollars in 2025 prices. The interval weighting regresses squared pair residuals on the holding interval and reweights. The hybrid adds single sales with living area, rooms and age. The smoothed row penalises the second differences of the index. Standard errors are the placebo variance of Arkhangelsky et al. (2021); the studentised column divides each placebo by its own pre-period fit before taking the dispersion, following the normalisation of Abadie et al. (2010), and the interval is a percentile of that distribution. The final row is the construction used in the main text, on its own donor pool.

A.2 How the donor pool is reached

Table A2 reports what each condition of Section 4 costs in counties. A county failing several is charged to the first it fails, so the rows decompose the pool rather than count violations. Every candidate is already annually reassessing, that condition having been imposed when the extraction pool was assembled.

Two of the five conditions deserve a reader's attention because they turn on the sorting variable rather than on the housing market. The anchor-coverage floor and the selection-imbalance test both ask about the parcels sorted by a prior sale of their own, and together they remove twenty-nine counties at each reappraisal, more than every other condition combined. Where a parcel is sorted by the median sale price of its census block group instead, neither quantity limits the design: coverage runs above nine tenths, and there is little unsorted stock left to be unrepresentative of. The counties they exclude are disproportionately large suburban ones in the Atlanta metropolitan area, whose trajectory resembles Mecklenburg's more closely than the coastal Florida counties that carry most of the fitted weight. The screens are reported here as the estimates were produced, under the sort those estimates use, and the counties they cost are recorded so that the cost is visible.

Table A2: Donor screening funnel.

	2011 reappraisal		2019 reappraisal	
	Dropped	Remaining	Dropped	Remaining
Candidate counties		78		78
No owner-occupancy source	2	76	2	76
Deed coverage low and rising	13	63	11	65
Anchor coverage below ten percent	19	44	19	46
Anchored parcels unrepresentative	10	34	10	36
Thin or empty quartile cells	1	33	2	34
Eligible pool		33		34
Minimum 25 sales in the thinnest cell	1	32	1	33

Notes. Counties removed by each condition of Section 4, applied in the order the eligibility file applies them; a county failing more than one is charged to the first. Candidates exclude Mecklenburg and are annually reassessing before the funnel begins. Anchor coverage is the share of the owner-occupied stock placed in a value quartile by a prior sale of its own, and the floor is a tenth; the imbalance test compares those sorted parcels with the unsorted on assessed value, improvement value and lot size, and requires the largest standardized difference below 0.25. The final row is the additional minimum-cell condition the price outcomes carry, which is why the price rows of Table 9 rest on fewer donors than the rate rows; one further county is lost in estimation for want of a complete series.

A.3 Counties admitted to the donor pool

Table A3: Counties admitted to the donor pool at the 2011 reappraisal

County	State	Coverage stability	Stock sorted (%)	Selection imbalance	Smallest cell	Owner-occupancy source
Mobile	AL	0.61	20.2	0.06	19	Occupancy field
Shelby	AL	0.57	48.0	0.23	26	Exemption code
Maricopa	AZ	0.78	57.7	0.05	1320	Occupancy field
Pima	AZ	0.71	53.6	0.05	231	Occupancy field
Pinal	AZ	0.75	41.7	0.06	98	Occupancy field
Yavapai	AZ	0.57	57.3	0.05	58	Occupancy field
Alachua	FL	0.70	48.0	0.12	37	Homestead roll
Bay	FL	0.63	34.4	0.14	18	Homestead roll
Broward	FL	0.58	62.0	0.16	344	Homestead roll
Charlotte	FL	0.52	53.6	0.03	52	Homestead roll
Citrus	FL	0.54	42.6	0.08	33	Homestead roll
Clay	FL	0.60	38.2	0.06	25	Homestead roll
Collier	FL	0.61	53.3	0.05	57	Homestead roll
Duval	FL	0.60	46.1	0.02	128	Homestead roll
Escambia	FL	0.65	42.2	0.08	44	Homestead roll
Flagler	FL	0.54	58.9	0.13	28	Homestead roll
Hernando	FL	0.55	53.6	0.05	52	Homestead roll
Highlands	FL	0.53	42.2	0.06	14	Homestead roll
Hillsborough	FL	0.55	46.8	0.04	223	Homestead roll
Indian River	FL	0.54	50.6	0.06	31	Homestead roll
Lake	FL	0.58	48.3	0.09	75	Homestead roll
Lee	FL	0.67	55.8	0.08	190	Homestead roll
Manatee	FL	0.58	44.5	0.10	62	Homestead roll
Marion	FL	0.55	41.6	0.04	61	Homestead roll
Martin	FL	0.51	56.1	0.06	41	Homestead roll
Miami-Dade	FL	0.60	50.1	0.08	162	Homestead roll
Okaloosa	FL	0.57	35.2	0.16	21	Homestead roll
Orange	FL	0.60	52.0	0.11	163	Homestead roll
Osceola	FL	0.65	46.2	0.04	44	Homestead roll
Palm Beach	FL	0.51	59.5	0.06	275	Homestead roll
Pasco	FL	0.50	46.5	0.21	129	Homestead roll
Pinellas	FL	0.47	52.7	0.04	196	Homestead roll
Polk	FL	0.59	44.2	0.10	119	Homestead roll
Santa Rosa	FL	0.57	44.9	0.11	34	Homestead roll
Sarasota	FL	0.58	51.5	0.09	111	Homestead roll
Seminole	FL	0.58	48.8	0.03	52	Homestead roll
St. Johns	FL	0.60	29.9	0.05	45	Homestead roll
St. Lucie	FL	0.67	52.6	0.16	95	Homestead roll
Volusia	FL	0.49	51.4	0.13	101	Occupancy field
Cobb	GA	0.86	59.4	0.17	27	Exemption code
Jefferson	KY	0.63	19.1	0.24	32	Exemption code
–	MA	0.87	40.1	0.09	19	Occupancy field
–	MA	0.89	39.8	0.01	12	Occupancy field
–	MA	0.86	44.1	0.03	17	Occupancy field
Ingham	MI	0.52	38.7	0.05	19	Occupancy field
Kalamazoo	MI	0.54	23.6	0.16	12	Occupancy field
Kent	MI	0.57	28.2	0.21	10	Occupancy field
Macomb	MI	0.37	42.1	0.14	76	Occupancy field
Oakland	MI	0.39	42.5	0.09	95	County GIS
Ottawa	MI	0.55	18.0	0.04	13	Occupancy field
Wayne	MI	0.74	30.2	0.16	161	Occupancy field
Arlington	VA	0.90	49.2	0.09	35	Occupancy field
Chesterfield	VA	0.73	42.1	0.10	81	Occupancy field
Fairfax	VA	0.89	50.4	0.03	232	Occupancy field
Loudoun	VA	0.94	50.2	0.13	100	Occupancy field
Prince William	VA	0.69	55.6	0.13	112	Occupancy field

Notes. Every county satisfying the five conditions of Section 3.1 at the 2011 reappraisal, with the value each condition took. Coverage stability is the county's pre-reappraisal sales volume as a share of its own best-covered four-year stretch. Stock sorted is the share of the owner-occupied universe placed in a value quartile by a prior sale. Selection imbalance is the largest standardized difference between sorted and unsorted parcels over assessed value, improvement value and lot size; the condition requires it below 0.25, and Mecklenburg's own value is 0.077. The smallest cell is the fewest sales in any quartile-quarter inside the estimation window. The owner-occupancy source records what identifies an owner-occupant, since the sources are not equally strong: a homestead roll is a statutory record, while the occupancy field is a coding in the transaction data. A dash marks a county whose name is not recorded in the extraction registry.

A.4 Synthetic control paths at the 2019 reappraisal

The 2019 reappraisal is not estimated in the main text, because the transaction extract ends in 2020 and leaves it two post-reappraisal quarters against four before it. Figure A1 shows the two outcomes that nonetheless pass the feasibility screen there, so that the excluded event is visible rather than merely asserted to be unusable. The pre-reappraisal fit is closer than at 2011, which is what six pre-period years rather than four buys; the post-period is too short to read a shape from, which is why it is excluded.

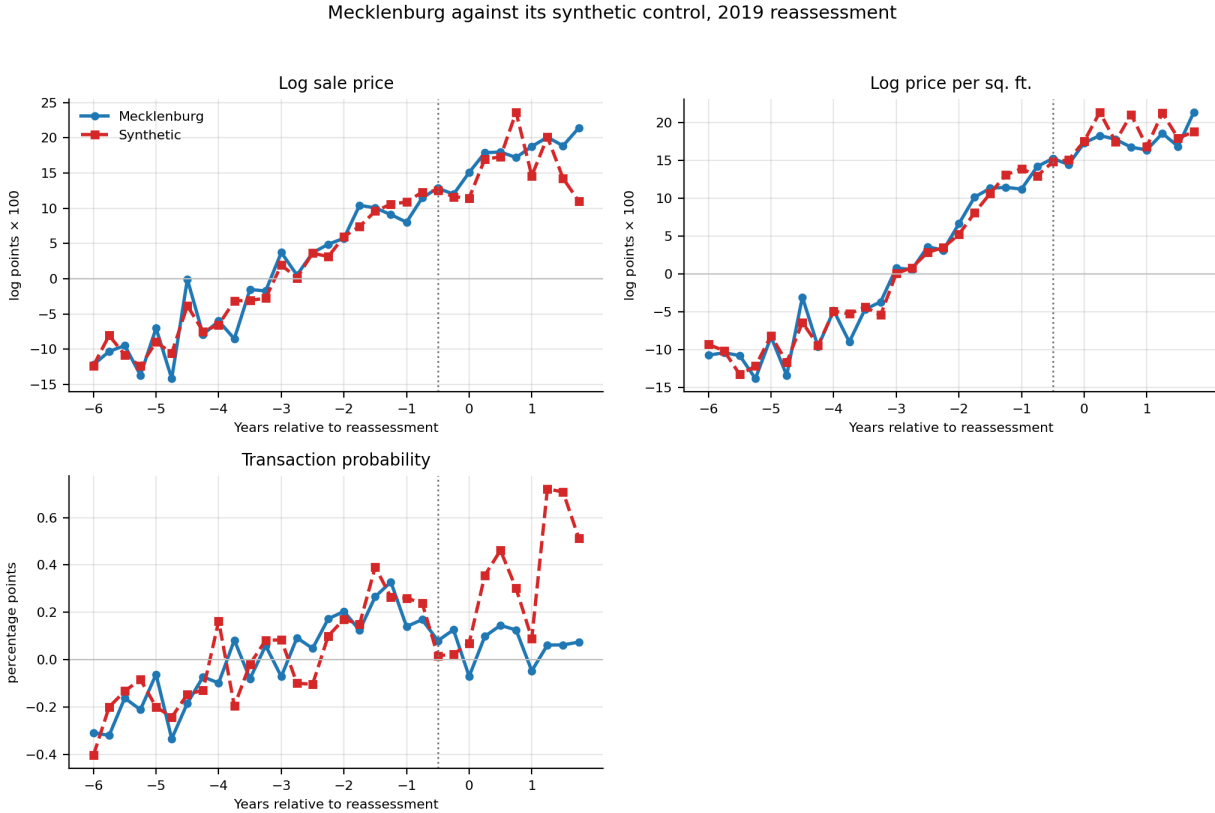


Figure A1: Mecklenburg against its synthetic control, 2019 reappraisal.

Notes. As Figure 2, for the 2019 reappraisal, over $\tau = -6$ to -1 before and $\tau = 0$ to $+1$ after. Only log sale price and log price per square foot pass the feasibility screen at this event.

Contributing donor counties, 2019 assessment (legend: weight on cheapest quartile / on the rest, %)

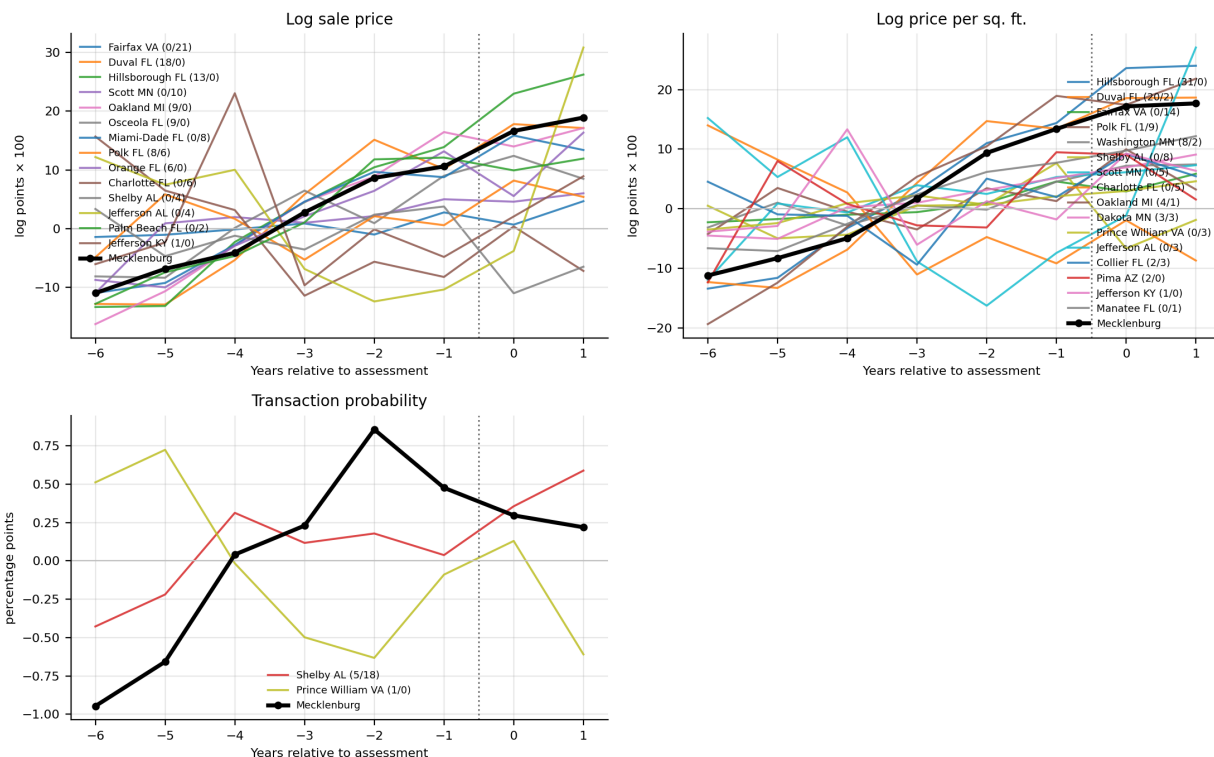


Figure A2: The contributing donor counties, 2019 reappraisal.

Notes. As Figure 2, for the 2019 reappraisal. More counties carry weight at this event than at 2011, the longer pre-period admitting a wider set.

A.5 Industry measures of assessment performance

The appraisal industry evaluates assessment quality with a standard set of statistics on the distribution of the assessment ratio, the assessor's value divided by the sale price. Table A4 reports them for Mecklenburg following McMillen and Singh (2020b). The value-weighted mean weights each ratio by its sale price, so it equals total assessed value divided by total sale price; when it falls below the arithmetic mean, expensive homes are being assessed at lower ratios. The price-related differential is the ratio of the two, and values above one indicate regressivity. The coefficient of dispersion measures uniformity as the average absolute deviation from the median ratio.

By these measures Mecklenburg appears close to compliant. The price-related differential is 1.022 in 2011 and 1.010 in 2019, both inside the industry's vertical-equity range of 0.98 to 1.03, and both coefficients of dispersion fall below the uniformity threshold of 15. Yet the same homes yield fitted slopes well below one. The two verdicts differ because the price-related differential is a ratio of two aggregates and is insensitive to curvature in the ratio distribution. Figure A3 shows the curvature directly: binned mean ratios fall from

roughly 1.20 to 1.02 across the 2011 price range, while the price-weighted aggregate barely moves because most assessed value sits in the middle of the distribution, where ratios are near one. The regressivity in Mecklenburg is therefore of a kind the standard industry diagnostic does not register.

Table A4: Assessment ratio statistics, Mecklenburg assessments of 2011 and 2019.

Reassessment	N	Central tendency			Dispersion		PRD
		Median	Mean	Value-weighted	COD	Value-weighted	
2011	5,230	1.027	1.046	1.024	13.2	12.8	1.022
2019	15,563	0.931	0.925	0.916	11.3	11.4	1.010
Pooled	20,793	0.948	0.955	0.938	12.4	12.0	1.019

Notes. Owner-occupied single-family Mecklenburg homes assessed and sold in the same year. The assessment ratio is the assessor's value divided by the sale price, so it is unit free and needs no price adjustment. The value-weighted mean weights each ratio by its sale price; the value-weighted coefficient of dispersion (COD) weights the absolute deviations the same way. The price-related differential (PRD) is the arithmetic mean divided by the value-weighted mean, with values above one indicating that expensive homes are assessed at lower ratios. Industry standards call for a COD below 15 and a PRD between 0.98 and 1.03. Outliers are removed with the nonparametric trim used by McMillen and Singh (2020b), which drops observations outside three interquartile ranges of the first and third quartiles of the log sale price, the log assessed value, and the assessment ratio.

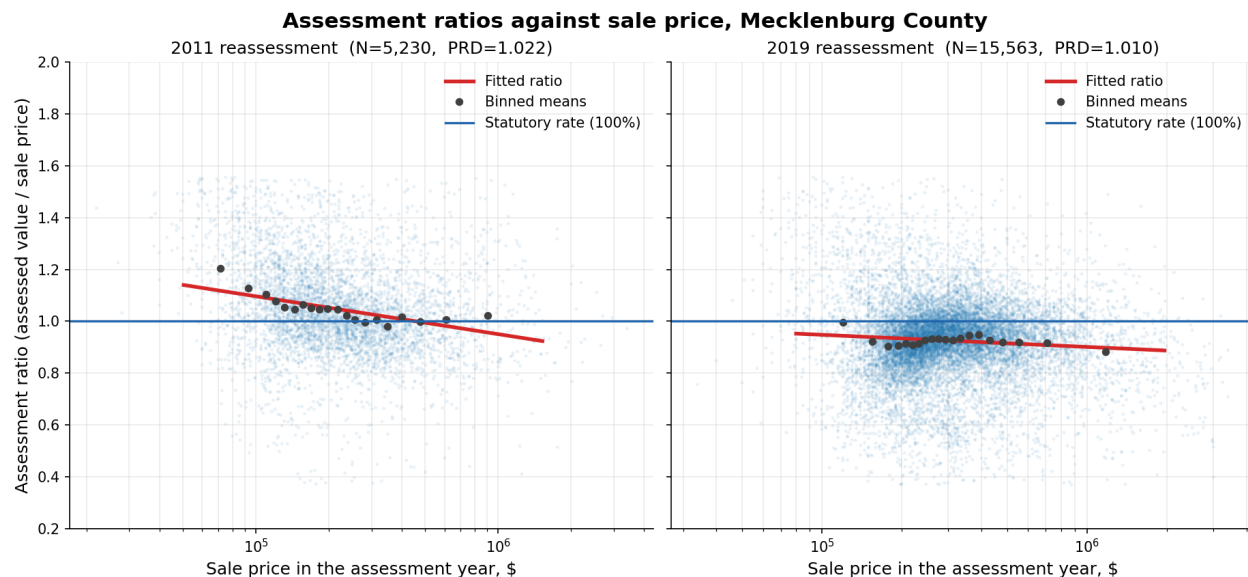


Figure A3: Assessment ratios against sale price, owner-occupied single-family Mecklenburg homes assessed and sold in the same year. The horizontal line marks North Carolina's statutory assessment rate of 100 percent of market value. The fitted line is a linear regression of the ratio on the log sale price; the binned means are averages within twenty equal-count bins of the sale price and trace the curvature the linear fit averages over.

A.6 Assessed value against sale price by reappraisal cycle

Figure 1 pools the 2011 and 2019 reappraisals. Figure A4 splits them, and the two cycles differ: the fitted slope is 0.777 in 2011 and 0.905 in 2019, so assessments tracked sale prices more closely in the later cycle even though both remain below one.

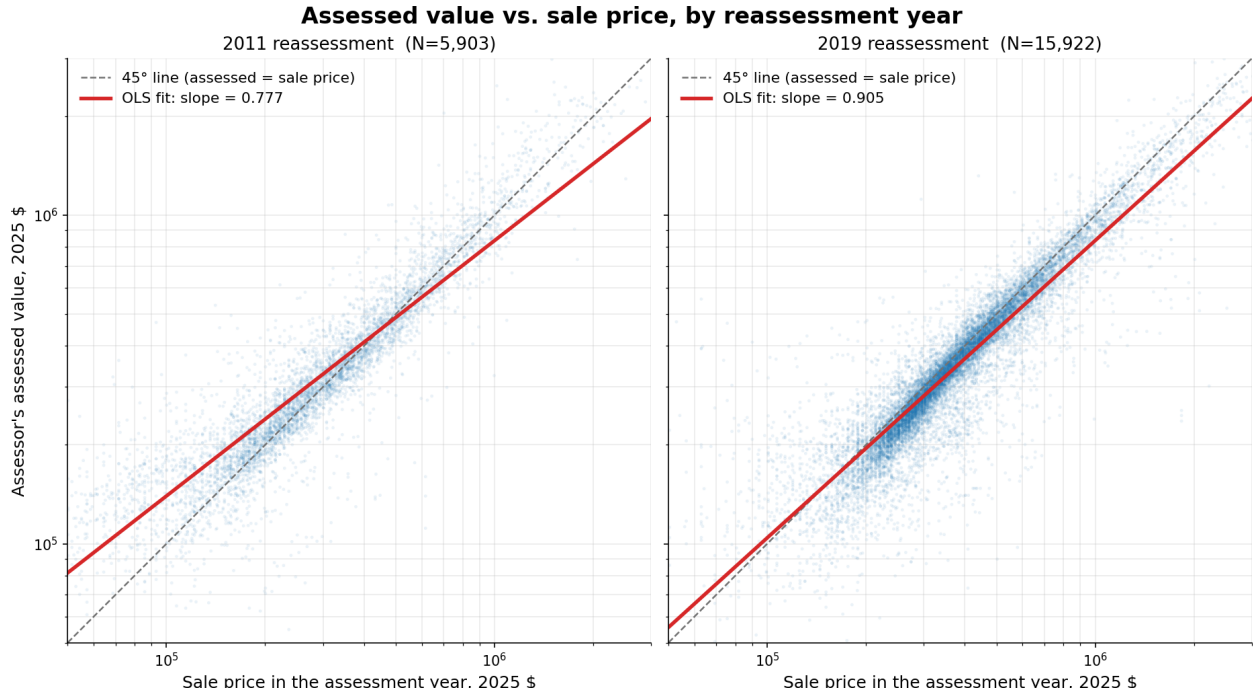


Figure A4: Assessed value against sale price by assessment year, owner-occupied single-family Mecklenburg homes assessed and sold in the same year. Both axes are in 2025 dollars. The dashed line marks equality between the assessed value and the sale price; the fitted slope lies below one in each cycle, which is the regressivity of the assessment.

A.7 Millage rates across the value distribution

Table 4 reports a single millage rate per year, since the rate is set at the jurisdiction level rather than by home value. Table A5 breaks the rate out by quartile. The four series are nearly identical within each year, so the gradient in tax-bill growth across quartiles reflects the reappraisal rather than any difference in statutory rates or in the municipalities cheaper homes sit in.

Table A5: Millage rate and change in the property tax bill by home-value quartile, Mecklenburg, 2012–2022.

	Millage (per \$100)				Change in tax bill (%)			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
2012	1.202	1.176	1.161	1.176	-2.30	-2.25	-2.21	-2.18
2013	1.222	1.193	1.187	1.218	1.95	2.03	2.03	1.55
2014	1.221	1.229	1.210	1.239	-0.03	1.48	1.49	1.83
2015	1.247	1.237	1.208	1.210	0.88	0.77	0.77	0.81
2016	1.260	1.231	1.213	1.230	0.28	0.20	0.32	0.46
2017	1.260	1.230	1.213	1.228	0.37	0.25	0.33	0.43
2018	1.276	1.245	1.228	1.241	1.31	1.26	1.23	1.20
2019	0.997	0.978	0.956	0.961	27.33	19.96	12.54	6.24
2020	1.005	0.985	0.960	0.964	1.45	1.03	0.74	0.78
2021	0.999	0.984	0.965	0.973	-0.25	0.04	0.61	0.96
2022	0.941	0.922	0.918	0.918	-4.29	-4.96	-3.95	-4.36

Notes. Owner-occupied Mecklenburg parcels by value quartile (Q1 cheapest, Q4 most expensive). Millage is the effective rate borne per \$100 of assessed value, the levy divided by the assessed value, averaged across parcels within the quartile; parcels in different municipalities face different combined rates, so the quartile means differ through municipal composition and through exemptions. The change in the tax bill is defined in equation (9). The horizontal rule marks the 2019 countywide reappraisal.

where the change in the property tax bill is

$$\Delta T_{it} = 100 \times \left(\frac{T_{it}}{T_{i,t-1}} - 1 \right), \quad (9)$$

with T_{it} the administrative tax levy on parcel i in year t . Dollar levels reported elsewhere are expressed in 2025 dollars, but the change is left in nominal terms because inflation is common to all parcels within a year and therefore drops out of any comparison across quartiles or out of any specification carrying year fixed effects.

A.8 The address-history link

Table A6 describes the data behind Section 5.6. Panel B is the coverage question and Panel C the selectivity question, which matters more: a link succeeding on large, new, expensive homes would describe those homes rather than the ones the exercise is about. Coverage is within a point of ninety-nine percent in every quartile, and the standardised differences between matched and unmatched homes run below a quarter, the threshold applied elsewhere in the paper. Matched homes are the slightly cheaper, smaller and older ones, so what selection there is runs toward the displaced rather than away from them.

Table A6: The address-history link, in summary.

<i>A. The address history</i>	
Source files scanned	384
Persons with a Mecklenburg address	4,197,509
Address records	12,730,048
Of which Mecklenburg addresses	6,699,523
Addresses per person, median	2.0
Addresses per person, mean	3.0
Records with an effective date	98.0%
Effective date, 10th percentile	1995
Effective date, median	2010
<i>B. Parcel-to-person link</i>	
Mecklenburg parcels in the assessor file	335,142
Matched, postcode and full street address	314,317
share of parcels	93.8%
Matched, postcode, house number and street	324,394
share of parcels	96.8%
Persons linked to a parcel	2,843,296
Parcels in the estimation panel	198,316
share matched	98.7%
share matched, Q1 (cheapest)	99.1%
share matched, Q2	99.0%
share matched, Q3	98.2%
share matched, Q4 (most expensive)	98.7%
<i>C. Matched against unmatched homes</i>	
Assessed value, matched	229,555
Assessed value, unmatched	275,702
Assessed value, standardised difference	-0.166
Living area, matched	2,274
Living area, unmatched	2,570
Living area, standardised difference	-0.243
Year built, matched	1988
Year built, unmatched	1991
Year built, standardised difference	-0.166
<i>D. Departures and destinations</i>	
Departures from a linked parcel, all years	2,311,052
parcels they leave	294,625
Departures in 2011 to 2016	289,455
on a parcel in the 2011 estimation sample	177,830
persons	168,542
parcels	82,733
destination in Mecklenburg	50.5%
destination in North Carolina	68.8%
destination matched to an assessor parcel	70.0%
destination carrying a block-group price	43.7%
destination carries a unit number	16.6%

Notes. Panel A covers the consumer address history described in Section 4; effective dates are the month an address took effect, and the percentiles are taken over all records held by people with a Mecklenburg address. Panel B reports both match keys: the first adds the pre-directional and the street type and understates coverage wherever a unit number is recorded on one side only, the second is the key used throughout. The estimation panel is the block-group-sorted sample of Section 4, described in 2018. Panel C compares matched with unmatched parcels in that sample on the assessor's own fields, in nominal dollars and square feet; a standardised difference is the gap over the pooled standard deviation. Panel D counts departures, defined as an address at a Mecklenburg parcel followed by a later address in the same person's history. A destination carries a block-group price when it matches a parcel the estimation sample prices, which is what the last three columns of Table 17 require.

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